

Seneschal | 保険エコシステムで閉じる相互警告

危険兆候を相互警告し、希少事象を自動運転学習データとして蓄積する。

保険会社・自治体・車両を結ぶ相互警告とデータ循環

A 想定主体

- 保険会社
- 自治体
- 自動車メーカー
- Tier1
- 歩行者・車両利用者
- Seneschal

B 保険会社への想定価値

- 事故リスク低減
- 損害査定額の検証
- 事故状況分析
- 責任判断の支援
- 契約者接点の強化



C 利用者への想定価値

- 危険接近の補助通知
- 警告・対応履歴
- 事故対応支援
- 安心な移動
- 安全サービスへの活用



D 自動運転開発側の想定価値

- 危険場面候補の蓄積
- 警告ログの蓄積
- 警告後の反応記録
- 警告ロジック改善
- 学習データ候補化



人と車の危険兆候を、路側機・クラウド経由で照合する

歩行者の情報 (Seneschal)

① 初期 | スマホ計測で補助警告

- 現在位置・進行方向
- 移動履歴・避難変化
- 速度変化からリスク推定
- 急停止・急加速
- 危険地点候補の記録
- 受信警告・反応履歴



② 長期 | 統合データ基盤に記録・再活用

- RSU・交通情報と連携
- 信号状態・歩行者検知
- 道路工事・気象情報
- 混雑・天候・視界

③ 予測用底向上 (RSU連携・死角補完)

スマホ×RSU×車両情報を照合し、予測しにくい歩行者行動の兆候を推定

- 急な横断の兆候
- 死角からの飛び出し候補
- 車載センサとの照合
- 希少情報の提供



情報として出さるもの

- 位置・方向・速度・加速度・履歴
- 過去の行動傾向・危険履歴
- 受信警告・反応記録・警告回数

RSU (路側機)



車が接近中！
止まって確認を

歩行者横断の可能性！
減速して注意

--> 情報送信 --> 警告配信

車両の情報 (AI対応ドラレコ/OBU)

① 初期 | AI対応ドラレコで検知・補助警告

- 接近車両・距離
- 障害物・飛び出し候補
- 車線逸脱・方向指示器
- 急制動・急加速
- 死角・視認度
- 時刻・位置



② 拡張 | ODD条件・車両状態を統合

- 車速・加速度
- 車線位置・ルート
- 制動入力・運転操作
- ADAS・自動運転AIの認識・判断
- OBU通信ログ・時刻同期
- 内開状態・磁場制

③ 学習用イベント候補 (切り出し)

ドラレコ映像と時刻同期して抽出 (品質評価前)

- 警告対象・飛び出し候補
- 警告から反応までの時間
- 遅延検出での差分
- 危険領域への進入
- 認識と実行の差分



1 現在を認識する
人と車の位置・進行方向・
速度を同一時刻・地点で
照合。



2 危険を予測する
人・車・RSUの情報を照合し、
予測しにくい行動の兆候と
衝突リスクを推定。



3 相互警告する
双方へ補助警告し、
危険回避を支援。

※車両制動・安全保証を代替しない

データを価値に変える | 保険・安全・自動運転AIへの循環 (提案価値)

① イベントデータ候補の収集

AI対応ドラレコ+Seneschal+RSU+OBUから危険イベント候補を抽出

- 映像・位置・速度・環境・車両状態
- AI認識・警告・反応を時刻同期
- 同乗・匿名化・品質評価後に蓄積

1

② 保険価値を検証

危険場面を分類し、保険サービスへの活用可能性をPoCで評価

- 事故リスク低減効果
- 損害査定額の可能性
- 事故状況の把握支援
- 契約者接点/事故対応

2

③ 安全価値を検証

相互警告による危険回避支援と事故リスク低減効果を検証

- 警告ロジック改善
- エリア別リスク評価
- 運用ルール最適化
- 交通安全への活用

3

④ 自動運転AI学習へ活用

評価・アノテーション済みイベントを認識・予測モデルの学習候補に

- 認識/予測性能の検証
- 希少事象の補充
- 安全評価への活用

4

⑤ 差別化へ変換

歩行者・路側・車載データの統合で、単一センサでは難く希少事象を収集

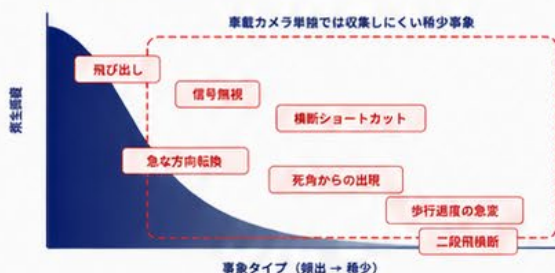
- 学習データの多様化
- 警告・予測の改善
- 保険・自治体・OEM連携

5

実装で得られるイベント候補

- 夜間・雨・雪・逆光
- 工事・死角・混雑
- 飛び出し・信号無視
- 横断ショートカット
- 進行方向・速度の急変
- 希少な危険・ヒヤリハット

ロングテール事象の整理 (歩行者行動に起因する希少事象)



同一時刻・地点の
人・車・路側情報を照合



複数情報の照合による
衝突リスク推定と
補助警告

データ源ごとの補充範囲 (定性比較)

データ源	主な提供範囲
A Seneschal	歩行者の位置・進行方向
A AI対応ドラレコ+OBU	車両・運転行動
B RSU	路側・交通情報
統合情報	単一センサの欠落を相互補充
定量評価:	



Federation | 街の“今”を捉え、人・交通・エネルギーを動かす

街の“今”を観測し、人の移動と地域需給を捉えて学習対象を広げる。

街を捉える分散観測・データ連携

A. 車載観測・ドラレコ		B. Seneschal (本人同意)		C. 道路・施設・運行システム	
IMG 道路画像	DVR 行列・渋滞情報	CAL カレンダー	HIS 移動履歴	CAM 道路カメラ・信号	PT 公共交通・運航
GPS GNSS・時間・方位	NID 走行軌跡	DOC 滞在位置	FIT 体調選択	FAC 駐車場空き	SOC SOC・蓄電残量
SPO 速度・加速度	MEN 急減速・介入ログ	STP 行動量	SWT 操作・効率性	CNG 充電設備空き	ENE 燃料・電力価格
3D 点群・物体情報	SEN センサー品質	TMR 移動時間情報	APP アプリ履歴	CON 工事・規制	OPS 営業所・運行D/B
ASN 車番・種別・入口	POI POI変化候補	LOC 次の予定	PAY 献付動向	EVE イベント・施設	
		STY 予約状況	ENV 充電選択数		

利用目的・価値観測・アクセス範囲を属性

地域ノード (路側センサー・RSU・エッジ/MEC) による時空間処理

画像処理	状態推定	時空間処理	学習候補抽出
IMG AI 命A*検知	CON 行人量	TSP 空間補間 (時空平均)	MAP レア事象スコア
SPO ナンフリーに検出	FLM 走行流量	TBK 軌跡異常検出予測	ASN 車種・POI変化候補
3D 点群・玉置検出	SPO 濃度	ZIR 定電電圧検出予測	LOC 異常音検出候補
ADR 暗黒・濃度異常検出	OCG 駐車場空き	ENV イベント見入度	ACT 歩歩検出候補
SEN 警告音検出	SOC 充電器空き	POI POI抽出・再検	QUL 品質劣化検出
QAL 品質劣化検出	SIG 交通状況	RIS 交通・充電開始予測	INP モデル更新候補
	SIG 工事・規制		

準リアルタイムStreet ViewとPOI更新情報

SAMPLE

渋谷区道玄坂
★ 22°C 晴れ
2025-05-10
12:03 SAMPLE

POI更新候補の生成	外部情報による属性補完	地図・ナビ更新への活用
NEW 新規出現	UPO 名称AI	MAP 再検出時空間画像の生成
DIS 閉店	NER 名称AI	POI POI更新情報の連携
REV 改装	POP 信用	NAV ナビ情報の更新
MER 統合判定	PRI 車庫情報	NAV 再度工数・経路の最適化
	VER 画像検証	
	VER 歩歩検出	
	VER 歩歩検出	

静的地図を進化し、経済活動の高め状態を表す「準リアルタイムStreet View」

学習候補への還流フロー: 観測 → 品質・権利確認 → 承認 → 反映

1 街の“今”を観測・構造化

- IMG 道路・入口・施設状態
- DVR 走行・軌跡・データ量
- RSU 地域ノード時空間処理
- SEN 画像・天候・環境状態
- POI 経路補正後の今の付与

2 学習候補抽出・品質検証

- MAP 道路・エッジ・異常検出
- ASN 車種・POI・入口候補
- POI 変化し歩歩の名称候補
- QUL 画像・検出・個人情報確認
- INP 有用度・信頼評価

3 検証・承認後に更新候補へ

- LVL 候補選択時の充足基準
- QUL 品質モデル適合度有用性
- LFR 権利設定・配信モデル検証
- LOG シミュレーション・再生
- SEC 協同名に反映保存

4 運行・サービスKPIを評価

- KPI 分野・Safety margin
- ESP 効率・POI指標
- EST 経済性指標・待ち時間
- ENE 運行効率化エネルギーコスト
- ENV 安全工数・フィードバック

観測 → 候補抽出 → 品質・権利確認 → 再現試験 → 承認 → 段階反映 → KPI評価

Time × Place × Behavior × Observed City State

Seneschalカレンダー (標準画面)

2025年5月10日 (土)

時間	名前	予定	手段	夜
7:00	起床			
9:00	起床			
11:00	起床	12:30 - 13:00	会議	
15:00		外出	15:30 - 16:00	
17:00		夕食	17:30 - 18:30	
19:00		帰宅	19:30 - 21:00	
21:00				

店舗・行列・混雑の観測例 (サンプル)

観測時刻: 2025-05-10 12:03:34

- 行列人数: 28人
- 待ち時間: 約25分
- 混雑度: 78%
- 空き駐車場: 12台
- 空き充電器: 3基

現在位置と経路マップ (サンプル)

目的地: Cofee & Bakery
到着予定: 12:28
残り時間: 25分
経路情報:
P 駐車場 12台
G 充電器 3基
GS (空) ¥168/L

① 時刻別の支援

OCR 朝出勤・日中帰	SHM 通勤経路予測
QAT 居の予定	ERR 混雑回避
ENV 歩行支援	QUL 天候補正
NIG 深夜・見守り	INF 料金

予定・移動条件・車の状態を一括制御/最適化/学習支援

② モビリティ需給調査

TAX タクシー待ち	CAR バス待ち	EVT イベント影響
SHA 人流交通量	GAR 空き台平均	GPO 駐車空き
BUL バス混雑	SBC シェアリング充足	OGA 駐車場の定V

需給変量を吸収し、運行・配電・供給補正の最適化に繋ぐ

③ 充電需給調整

SOC 車両SOC	PRC 需要予測	PER 余剰制御
NEE 充電必要回数	CNG 充電設備空き	QSA 充電も必要
GOO 充電の遅延	POW POW(充電率)	QSA 充電補助・ピーク

交通需要と電力需給連携一時供給で予測し、充電配分を最適

自治体課題に対する活用案

課題	活用案
駅前混雑	人流把握・分散誘導
イベント時の渋滞	需要予測・時間配車
観光地渋滞	経路分散・交通流制御
タクシー不足	需要予測・車両配置
高齢者移動	オンデマンド交通支援
交通空白地	路線・車両最適設計
駐車場不足	空き情報・駐車誘導
充電器不足	空き情報・充電誘導
地域交通赤字	需要予測・運行効率化
電力ピーク	充電時刻の平準化

事業開発・開発環境

Federation

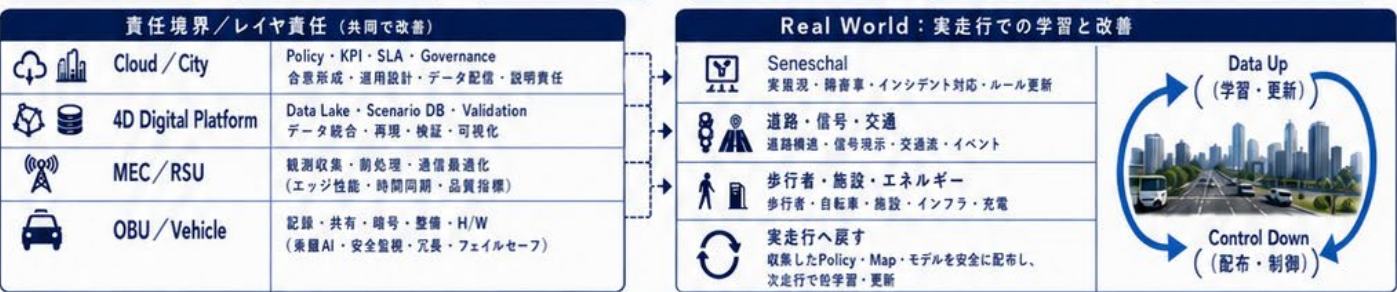
人流・物流・医療・行政を地域Nodeと共通運行基盤で束ね、地域に必要なサービスを一つの自動運転ループで回す。

<https://federation.base44.app>

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Architecture | 実走行をモデル更新へ返す開発基盤

高価値事象を検証・モデル更新へ返し、安全性とODDを継続的に高める。



街で得た高価値事象を検証・更新へ返し、次の自治体サービス実装を安全に拡張する。

Seneschal | 人口減少を反転する地域自動運転サービス

高めた自動運転基盤を24Hの地域生活サービスへ展開し、人口減少下の生活制約を解く。

自治体が放置できない3つの生活制約



少子化

女性就業が標準化しても、
送迎・見守りが家庭に残り、
就業継続・出生率を阻害



高齢化

1対1訪問と固定デイケアが、
介護人材と高齢者の
外出機会を奪う



孤独！

単身世帯が増える一方、
住宅地に夜の居場所と
安全な移動交通がない

日中 | 移動式介護サービス



介護職員が複数人を同時に見守る



共通目的地ごとに集まり、自由に外出



人材不足を緩和し「移動の自由」を回復

放課後～夜 | 移動式学童



保育者が運転せず、子どもを見守る



学校→習い事→自宅まで連続して預かる



就業・第2子／第3子・転入定住を支援

夜～深夜3時 | 移動式バー



バーテンダーが接客しながら住宅地へ運行



帰宅時間を、安価で日常的な交通時間へ



通勤・飲酒運転を減らし、夜間消費を増進へ



08:00-15:00

介護



15:00-21:00

学童



21:00-03:00

バー



03:00-08:00

充電・清掃・整備



サービスを替えても、車両・運行・予約・Node・データ基盤は共用

三つの専用事業ではない。三つのサービスコンテナで、一つの自動運転基盤の需要の谷を埋める。

実行フロー（24時間を支える一つの運営基盤）



自治体 | 施設投資を抑え、
複数改革を一つの運行基盤で実行

- ☑ 財政健全・維持費を抑制
- ☑ 福祉・教育・地域振興を同時推進
- ☑ データで施策最適化



住民・事業者 | 移動とサービスを一体化し、
待ち時間を生産資産を創造

- ☑ 移動時間と待ち時間の短縮
- ☑ 固定費・運営の最適化
- ☑ 利用機会と・地域連携の活性化



学習データ | 時間帯別需要・稼働・
運行・介入・エネルギーを基盤取得

- ☑ サービス改善を高効率化
- ☑ 需要予測・車両最適化を高度化
- ☑ エネルギーと充電計画を最適化

サービス利用者メリット



子育て世帯 | 仕事が生育から遠ざかずに済む



高齢者 | 介護を受けても行き先を自分で選べる



働く人 | 一次移動が、日常的につながる

自治体メリット



人口 | 共働き・第2子・第3子・若年世帯の転入定住



介護 | 人人材確保支援・重度化予防



安全 | 通勤・飲酒運転削減・夜間交通確保



財政 | 施設建物を抑え、基金を24時間活用

自治体へ車両を売るのではなく、人口減少を反転する生活サービス運営パッケージとして導入



Federation | 自治体共同事業で地域運行を束ねる

地域の移動・物流を共通運行へ束ね、自治体サービスを面的に維持する。

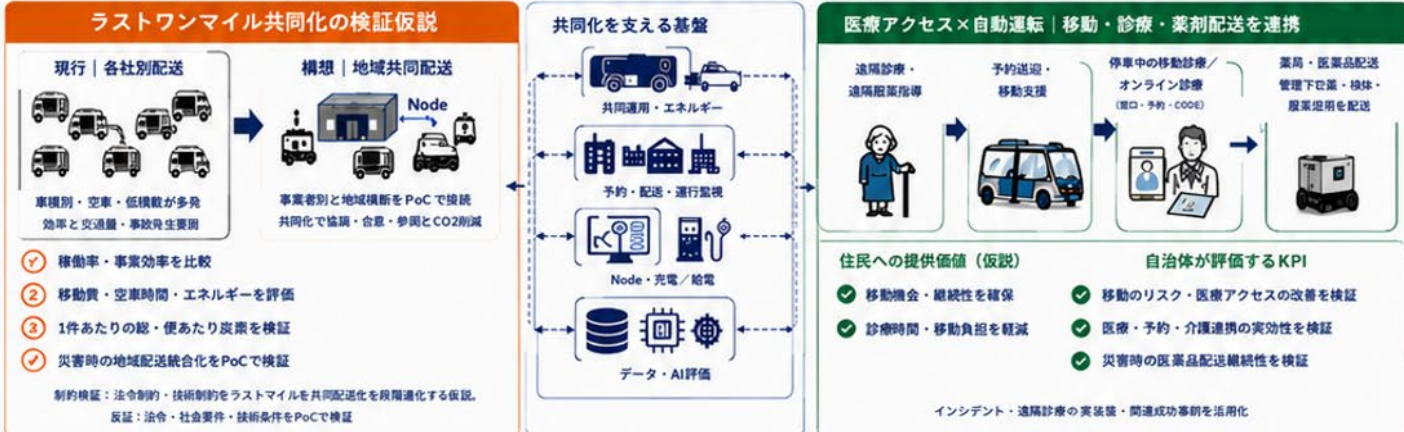


- 1 貨客混載（制度・安全要件前提）**
客室と荷室を区分・部止し、旅客・荷物双方の同意、運行範囲、地域協定を担保
- 2 共同配送EV（複数荷主）**
複数荷主の荷物を共同配送するEV
- 3 自動配送ロボット**
拠点から受取拠点までのラスト区間を自動走行・配送
- 4 移動行政サービス（停車型）**
停車拠点を巡回し、窓口・相談機能を提供



公開技術と独自構想を、認知・学習・検証・運用更新の同じループで接続

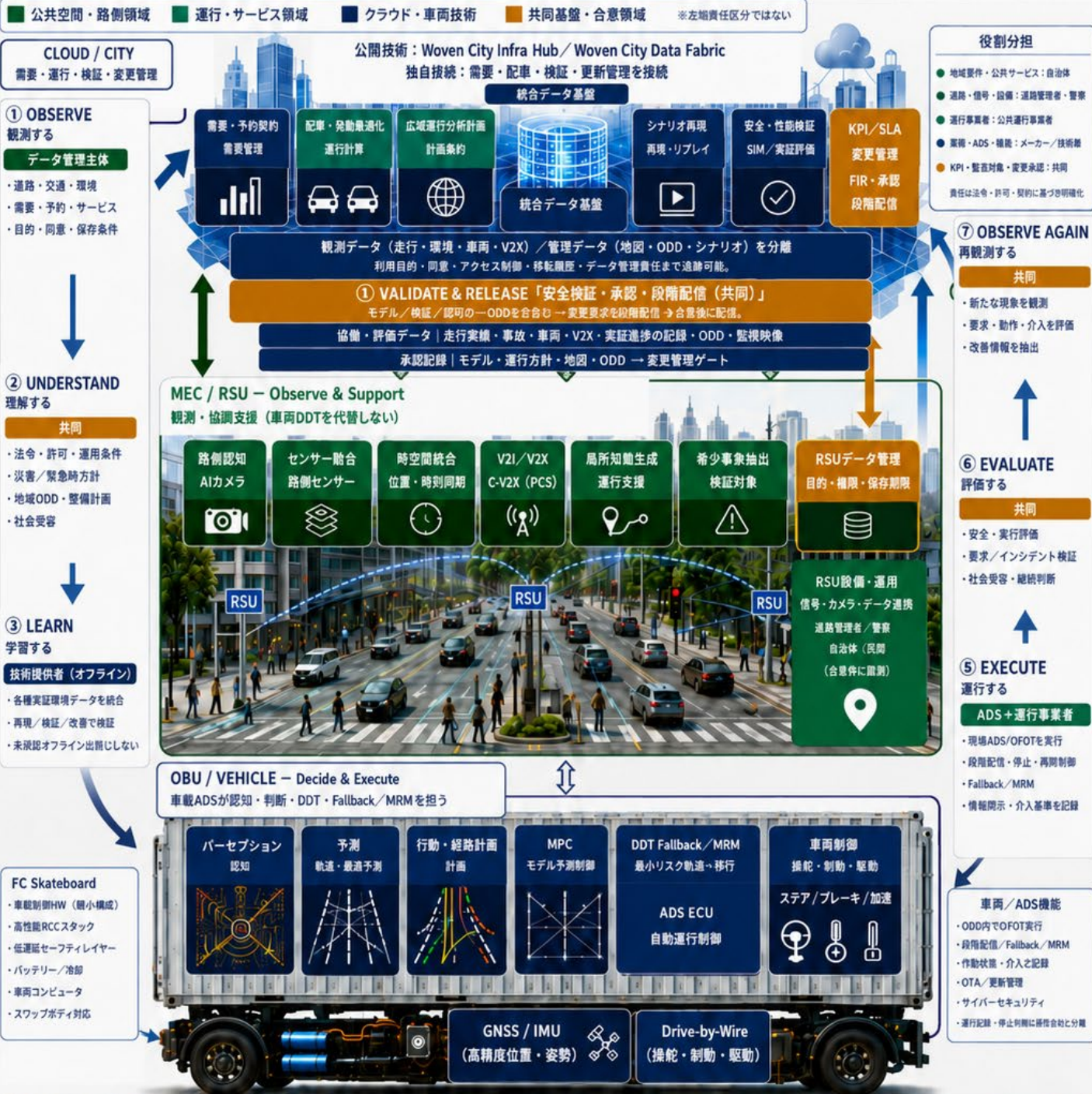
公開情報で接続できる技術基盤	独自構想で接続できる機能	事業・運用への接続（未実証）
- Woven City AI/IoT/Urban Engine・Woven City Behavior AI 統合・連携・評価フレームによる都市基盤連携 - Woven City Integrated A2EN2 System 観測と風光も入力、人・モビリティ協調、インフラの統合基盤 - Woven City Infra Hub・Woven City Data Fabric 都市データの統合と共有のハブ、プラットフォーム基盤 - Automated driving・Active Learning Loop 走行データから学習を促進し、道路・知識を両立 - Woven City Digital KAIZEN Platform 水素利用によるエネルギー・環境のエンジン・検証 - Arena SDK / Arena Tools / Arena Data 開発ツールの一貫・実行・データ・認証の連携基盤 - Woven City eVPP 仮想検証・シミュレーション・デジタル検証基盤 - Woven City Robot Platform 自動配送ロボット・移動型ロボットの運用基盤	- 利用意図・需要 - 相互警告 - 都市データ連携 - 都市少事象学習 - 可視化・検証 - 意思決定・最適 - エネルギー連携 - 社会実装 設計・学習の改善ループを構成	Federation RSU・OBU・外部データを運行制御へ接続 地域全体でのPCC機能、共同配送、運行最適化を支援 車両管理 保守・整備・故障検知を統合管理 整備履歴・点検記録・稼働状況を、資産管理系などと連携 データ連携 自治体・企業・住民とのAPI・Tier1 行政、物流、医療、インフラと統合、情報共有基盤を構築 シナリオ学習 周辺ODDから希少事象を抽出 反復データで知識を蓄積し、適応を強化 Digital Fabric / Ground（基盤） End-to-Endデータ・統合・セキュリティの基盤 Data Up / Control Down（制御） 収集・可視化・運用層への反映方法を構築 PPA・Node（電源） 水素・電力・通信・熱源の連携 運用サービス連携（環境） 水素・整備・充電・物流・防災などを運行基盤で統合



共同配送で車両・拠点の稼働率を高め、医療・行政サービスへのアクセス維持へ展開する仮説。

Architecture | 自治体共同事業の機能配置

自治体と民間の責任を分け、共同運行を継続改善できる実装構造にする。



Seneschal | MaaSと自動運転事業の違い

共同運行の収益性は、積載率・24H稼働・燃料費・共通基盤で決まる。

現行 MaaS | 人が運転する自動サービス

支配 KPI	支配原価
需要平準化率 40 %	人件費 45 %
稼働率 30 %	燃料費 40 %
稼働者 20 %	車両維持費 10 %
収益事業率 10 %	その他 5 %

運転手と個別需要の確保が事業を支配

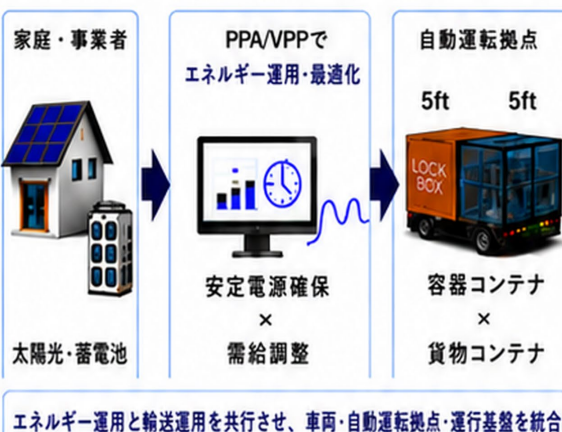
自動運転事業 | 共通基盤で24H運転運用

支配 KPI	支配原価
稼働率 50 %	車両費用 45 %
需要平準化率 25 %	OS費 25 %
24H稼働率 20 %	車両維持費 20 %
収益事業率 10 %	その他 10 %

人件費低減故は、稼働・連続稼働・エネルギー・OSが事業を支配

PPAサービスを展開する事によって自動運転事業の事業基盤を構築する

PPA/VPPサービス = 自動運転事業の事業基盤



自動運転車の配車 (自宅低圧盤を時間借り)



自治体のメリット

遊休施設や土地を有効利用する事が出来、税金依存財政を上げながら、人を呼び寄せる事が出来る

- 子育て世帯の転入・定着で人口減少を反転
- 宇宙・介護・物流・行政コストの一体合理化し
- 遊休地を活用し、災害時は電源・輸送拠点へ

利用者のメリット

光熱費を下げながら、待ち時間なく、いつでも利用する事が出来、移動時間の有効活用が可能になる

- 遊休時間が95%を占める車の保有が不要
- 駐車場も削減でき、好きな立寄り先で良い
- 場所を探すだけで、エネルギー稼働を経済不要

事業者のメリット

自動運転車両に必要なとなる拠点を構築し運用に使用されない運用費用の運用を変える事が出来る

- エネルギーコストを最小化してネット化
- 遊休時間を最適化する事で稼働率を拡大する
- 土地代を低くのと同時に、満車の運用を構築

自動運転事業盤となる拠点バリエーション

ビル屋上拠点 | ドローン・水素充填



物流拠点 | 積載中・水素充填



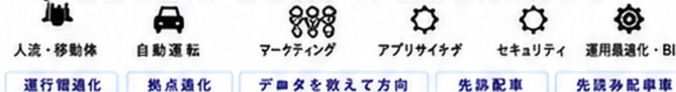
夜間輸送 | 書籍返却・車両清掃・水素充填



星明 (コンテナショップ)

Dynamic Service (サービスの動的化)

必要な時に、必要な所へ、必要なだけ届ける



Dynamic Hub (拠点の動的化)

自動運転によって、社会のムリ・ムダ・ムラを削減



実証運用PoC

Seneschal

MaaSの個別移動最適化ではなく、稼働率・24H拠点・共同運行・分散PPAを束ね、自動運転との共同運行を持続事業に変える。



<https://seneschal.base44.app>

Federation | 自動運転事業成立の車両・拠点要件

成立条件を、FCスケートボードと自動運転拠点の要件へ落とす。

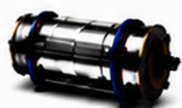
車両要件 FCスケートボード

- 重量化が不可能なCEやPCEVにて、車両重量を軽減
- 自動運転に必要なバッテリー容量による航続距離を確保
- コンテナ積載による車両選定上にて、重載貨物輸送を実現
- 東上水負荷量による車両選定基準で、積載量を担保
- 後輪操舵による後輪操舵によって、コンテナのコストを最小化
- 低温即発熱H1循環によって、温度での水素供給最適化

拠点要件 効率的な水素供給

- 車両側へのH+低温MH供給により、装着供給容量の増減
- 水素量(LH₂)、水力発電量した電解水による水素供給
- 水素SNMEにより、外力を活用した水素の在庫力量の変化
- 混合水素の移動を活用したHPC高圧自釐化による低圧増幅
- 混合水素の供給を活用したHTSシステムを向一効率的・水素消費
- HTSコイル等による運転制御にて、電力節電力流量に全対応

1 車載モル (水素循環)



- ・安定・メンテナンスに、またの低減圧をだけで水素を循環
- ・チタン製のトルクジェネカターの長期循環・長寿命化

2 MS-SOFC/PC-SOFC (PCFC)



- ・高効率・高信頼性により、水素利用の低温自循環
- ・本体の小型化・出力密度の向上

3 メインシャーシー構造 H+低温MH



- ・低温貯蔵の定性で水素貯蔵重量の軽減と減量
- ・安全・安定で高信頼高く、長期貯蔵が容易実現

4 クロスメンバー構造 HTSコイル&LH₂タンク



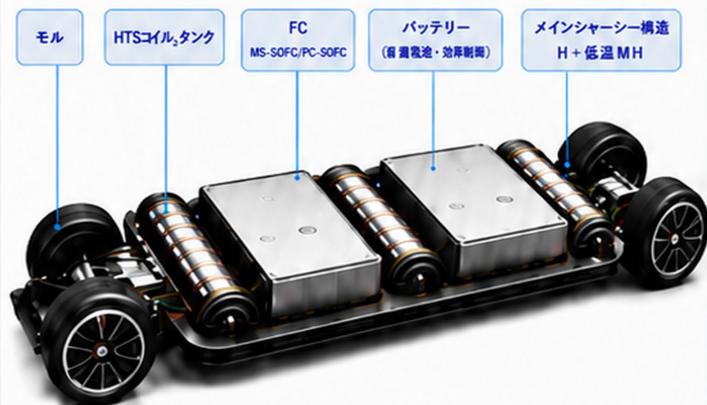
- ・連続利用によるエネルギー効率の向上
- ・低温利用による歩留りをもいクルマ作り

5 LH₂冷却HPC



- ・高密度での損傷信頼部品向上による10min以上を
- ・低温利用による高/多アーチファチャの最適

FCスケートボード



移動時間の有効活用を実現する為に必要な
“辞わせないクルマ作り”

HTS統合車両運動制御

HTS-IWM (統合モーター)



- 最高出力: 300 kW 以上
- 最大トルク: 1,200 Nm 以上
- 効率: 97% 以上
- 定格電圧: 800 Vdc 以上
- 出力密度: 10 kW/kg 以上
- 連続トルク密度: 25 Nm/kg 以上

HTS-バウス



- 定格: 50 kN 以下
- 作動電圧: 0.8 kV
- 転送電圧: ステップ制御
- 出力密度: 6 kW/kg 以上
- 応答時間: 1 ms 以下
- 連続耐久: 2,000 h 以上

HTS-ブレーキ



- 定格: 50 m 以下
- 制動減速度: 2.1~2.5
- 常用温度域: 高頻度ブレーキ制御
- 熱容量: 15 kJ 以上
- 応答時間: 10 ms 以下

HTS-インバーター (DC/AC・BMS)



- 出力: 300 kW rated
- 出力密度: 200 kW/L 以上
- 効率: 98% 以上
- DCバス: 800~1,200 Vクラス
- 切替周波数: 20 kHz 以上
- 絶縁耐圧: 4.0 kVrms 以上

HTS-エアサス・CFRPサス



- HTS化とCFRP化でエアサス化
- HTS・減震力: 5,000 N 以上
- 耐荷重: 10 t 級
- 応答時間: 150 ms 以下
- 応答時間: 1.5 kW 以下
- 軽量化率: 20% 以上

HTS-CFRPハイパフォタイヤ



- 外径: 1,200 mmクラス
- 耐荷重: 6 t 級
- 低発熱設計: -30~45°C
- グリップ性能: 2.0 g 以上
- 転がり抵抗: 20% 低減
- 耐久性: 3~5万 km

進め方

1 要求分析・コンセプト設計 (~2026)



- ・ステークホルダー調査 / ミッション定義整理
- ・ユースケース・運用設計・COOD/加減速
- ・要求分析・トレードオフ・技術検証計画
- ・システム構想・実証アーキテクチャ定義

2 要素技術開発・設計 (2026~2028)



- ・要素アーキテクチャ / 材質アーキテクチャ設計
- ・systemモデル化・シミュレーション / HMI・制御構築
- ・評価計画・試験計画・試作設計計画

3 拠点向けに実証運用 (2028~2030)



- ・実証運用拠点による本実証
- ・性能評価・改善検証評価
- ・安全性・環境性・拠点最適化
- ・集約化・運用効率整備

4 拠点大幅に最適化 (2030~)

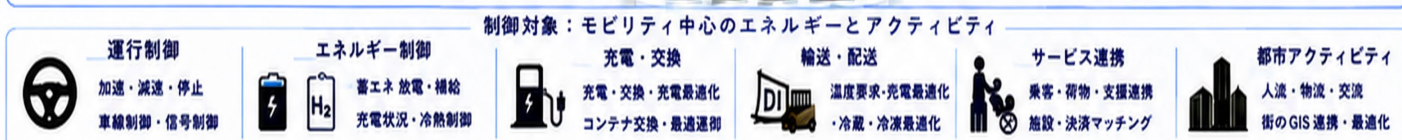
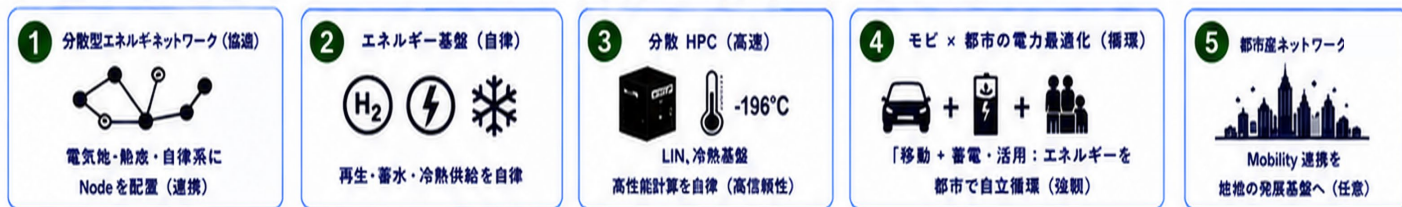


- ・コンテナ運用量最適化
- ・運用量・保全最適化
- ・運搬・コスト最適設計
- ・運用先・サービス & オペレーション整備

HT-PEFCによる本体小型化と軽減CO₂低減を比較検証

Architecture | 自治体実装を支えるエネルギー・演算基盤

車両・拠点をエネルギー・演算・通信でつなぎ、街事業開発へ拡張する。



AD-MaaS Business Logic

Seneschal: <https://seneschal.base44.app> ... Demand-generation implementation example
Federation: <https://federation.base44.app> ... Urban-operation packaging study framework

The success of AD-MaaS is determined not by autonomous driving technology itself, but by a business structure that converts operation into profit. As driver labor cost decreases through automation, the dominant cost factor converges toward energy cost, while vehicle utilization is required to recover CAPEX and load factor is required to increase revenue per trip. Seneschal smooths demand fluctuations by matching customer availability with mobility and services, while motion-sickness-free vehicle control transforms travel time into work, rest, and service value.

Because vehicle-grade quality requirements are strict, next-generation vehicle technologies should first be developed for stationary applications, then installed into service vehicles as they become ready. Service vehicles can absorb CAPEX through high utilization and high load factor, enabling a DevOps-style development model unlike conventional mass-production vehicles. Federation increases vehicle utilization through time-based service exchange and ridesharing, and increases load factor through passenger-freight integration and collaborative logistics, while planned mobility schedules optimize PPA energy management and enable AVP-integrated fueling to establish a carbon-neutral energy platform for autonomous mobility. AD-MaaS is an urban operating system that controls demand, vehicles, energy, and infrastructure as one system to convert utilization, load factor, and energy cost into profit.

PPA Service for Autonomous Mobility Foundation

As autonomous driving reduces driver labor cost, energy becomes the dominant operating-cost factor. The PPA service therefore establishes structural energy-cost competitiveness for AD-MaaS through distributed energy infrastructure.

Corporate PPA services support Scope 3 decarbonization while establishing distributed eVTOL operating hubs. Idle corporate vehicles are shared for employee commuting to improve fleet utilization and reduce Scope 3 transportation emissions. Residential trigeneration services establish distributed Micro Communter hubs, providing the node density required for daily autonomous mobility services.

Blockchain-enabled P2P energy and mobility sharing connects corporate and residential nodes into a mobility ecosystem, while renewable electricity, water-pipeline micro-hydropower, magnetic refrigeration, hydrogen production, superconducting power cables, liquid-hydrogen pipelines, and liquid-hydrogen-cooled HPC establish the infrastructure foundation for infrastructure-coordinated autonomy and continuous ODD expansion.

Infrastructure Layer	Control Variable	Target	Business Purpose
Renewable PPA	Renewable supply ratio	Maximize	Low-cost renewable electricity
Water Infrastructure	Micro-hydropower output	Maximize	Distributed baseload generation
Hydrogen Production	Electrolysis utilization	Maximize	Hydrogen production
Cooling	Magnetic refrigeration efficiency	Maximize	LH ₂ production & HPC cooling
Energy Network	Superconducting cable utilization	≥80%	Inter-node power sharing
Hydrogen Network	LH ₂ pipeline utilization	≥70%	Hydrogen distribution
HPC Node	AI compute availability	≥99.9%	Infrastructure AI computing
Infrastructure Coordination	Infrastructure coverage	Expand	Vehicle–infrastructure cooperation
Autonomous Mobility	ODD coverage	Expand	City-wide autonomous operation

AVP-to-MaaS Execution

This concept begins with an existing OEM customer use case: enabling refueling to be completed while the user is shopping. → A: time control
By integrating infrastructure-coordinated automated valet parking (AVP) with fueling, vehicles autonomously park, refuel, and return within dwell time, eliminating dedicated refueling stops. → B: time + location control
As autonomous driving progresses, parking demand shifts, creating underutilized parking spaces repurposed as operational nodes. By leveraging these assets, dual-use infrastructure enables daytime refueling for private users and nighttime fleet operations. → C: time allocation control
Liquid-hydrogen-cooled HPC nodes provide compute capacity for infrastructure-coordinated autonomy. Any ODD expansion remains subject to scenario coverage, V&V evidence, a validated safety case, and regulatory approval. Applied to large-scale parking operators, operational nodes exchange modular service containers according to time-of-day demand, balancing utilization across mobility, logistics, retail, and public services. This transforms Node into a MaaS foundation that mitigates demand fluctuations and increases asset utilization. → D: operational control
Extending this logic to public roads evolves the system into a city-wide mobility platform. → E: flow control
Through autonomous driving, vehicles synchronize in time and space, forming organized carrier operations rather than independent movements. → F/G: synchronization / platoon control
This expansion to public roads and coordinated carrier systems enables a new transportation system. Because this system is built on coordinated flows and shared infrastructure, Autonomous mobility expands from vehicle operation into integrated mobility and infrastructure coordination. → H: system-level control

Logic	Laver	KPI	Formula	Value	Unit
A	Demand	Demand generation rate	generated demand ÷ total potential demand	≥0.30	ratio
A	Demand	Conversion rate	booking ÷ request	≥0.23	ratio
B	Demand	Matching rate	matched ÷ request	≥0.90	ratio
D	Demand	Trip request rate	executed trips ÷ matched	≥0.95	ratio
D	Operation	Dispatch success rate	completed dispatch ÷ dispatch request	≥0.95	ratio
D	Operation	Dispatch response time	total dispatch response time ÷ dispatches	≤60	sec
D	Operation	Utilization	operating time ÷ total time	≥0.75	ratio
D	Operation	Fleet idle ratio	idle vehicles ÷ total vehicles	≤0.25	ratio
F	Vehicle	Latency	response time	≤1	sec
F	Vehicle	Control delay	actuation delay	≤20	ms
F	Vehicle	Planning latency	planning delay	≤100	ms
G	Vehicle	Control stability	spacing variance	≤5	%
E	Infrastructure	Throughput	vehicles ÷ time	≥40	veh/h
C	Infrastructure	Node utilization	used nodes ÷ total nodes	≥0.80	ratio
F	Infrastructure	Node latency	node delay	≤50	ms
H	Infrastructure	Energy cost per trip	energy cost ÷ trips	≤¥500	JPY/trip

Autonomous Mobility Business Execution Architecture

— Driving autonomous mobility business development through AD/ADAS system foundation —

Autonomous Mobility Business Definition

This architecture is defined as a control system where KPI, system behavior, and business outcomes are directly connected. This architecture directly connects KPI, system behavior, and business outcomes as a single control system. The objective is to build an AI-enabled autonomous mobility business architecture that improves utilization, load factor, and unit economics.

This is achieved by integrating mobility, logistics, energy, and urban services through a common operating architecture, while structurally improving:

- utilization
- load factor
- unit economics

This proposal defines a business architecture that makes this system executable, economically grounded, and deployable.

Autonomous Mobility Business Execution

This proposal defines a phased development and deployment plan for autonomous mobility business execution. Today's mobility systems fail to scale because demand, fleet, energy, and operations are still managed separately. This architecture integrates these elements into a unified autonomous mobility business structure, using autonomy as the foundation and AI as the means to operate autonomous mobility at business scale. System performance directly translates into revenue through utilization, load factor, and energy efficiency:

- Demand creation synchronized with dispatch
- Utilization maximization through modular vehicle operation
- Load factor optimization through dynamic vehicle composition
- Energy cost reduction through on-site hydrogen production
- Real-time operational control through distributed node coordination and AI-enabled optimization

Autonomous Mobility Business Architecture

Objective: To define an executable autonomous mobility business structure where AI-driven control logic is applied through autonomous driving to integrate demand, operations, vehicles, and infrastructure.

This architecture shifts mobility from fragmented services into a unified operating system where utilization, load factor, and energy efficiency are structurally optimized.

AI is applied as controlled logic across four layers.

1. Demand Orchestration

2030: Coordinated AI Assistant

A calendar-based interface connects user schedules with service availability, enabling demand generation through matching and reservation-linked mobility.

2035: Trusted AI Partner

A 4D digital platform integrated with map infrastructure synchronizes city-state and user behavior in real time, enabling continuous demand generation and capture to maximize load factor.

2. Operation Control

2030: Integrated Control

Energy management and dispatch are coordinated through decision logic that can later be augmented by AI, enabling stable and efficient operations under constrained conditions.

2035: Synchronized Control

Real-time synchronization of in-car and out-car hydrogen and electricity flows creates the control foundation for AI-assisted optimization, while structurally improving business viability.

3. Fleet Vehicle

2030: Swap Carrier

A modular vehicle structure enables continuous operation through container exchange, maximizing utilization.

2035: Composed Carrier

Vehicles dynamically couple and reconfigure during operation, enabling real-time redistribution of passengers and cargo to maximize load factor.

4. Distributed Infrastructure

2030: Urban Node

Distributed hubs provide localized energy and compute support, enabling operational independence and cost efficiency.

2035: Node Mesh

Distributed infrastructure is synchronized as a network, enabling real-time coordination between vehicles, energy, and computation.

This architecture enables a system where mobility, energy, computing, and AI-based decision-making operate as a unified layer, enabling scalable autonomous mobility business operation.

AI-Enabled Urban Mobility Roadmap

Autonomous mobility becomes deployable only when a production-ready driving stack, closed-loop learning, auditable safety, fleet operations and service economics are developed as one system. This roadmap first establishes that industrial baseline, then extends AI control from vehicles to distributed infrastructure and city-scale operations.

Industry Baseline & Capability Build-Up

Capability	Industry Baseline	Roadmap Focus
Product / Release Governance	Common requirements, configuration control, release evidence and change management across vehicles, sites and ODDs	Stage deployment through safety, service-readiness, availability, recovery-time and unit-economics gates.
Driver / Sensor Platform	Multimodal camera–lidar–radar perception, overlapping coverage, redundancy and deployment across multiple vehicle classes.	Common driving stack, hardware abstraction, calibration, fail-operational design and rapid vehicle onboarding.
Data / Active Learning	Fleet-scale data collection, anonymization, annotation, rare-event mining and uncertainty-based active learning.	Closed loop: detect → anonymize → select → label → train → validate → staged release.
Simulation / V&V	Real-event replay, scenario variation, closed-course and public-road tests, synthetic simulation and automated regression.	ODD-specific scenario coverage, acceptance criteria, critical-event recall, safety evidence and validation gates.
Safety / Remote Support	ODD safety cases, independent evidence, fallback/MRM, remote monitoring and contextual assistance with clear authority boundaries.	Claims, evidence, escalation, incident response, cybersecurity, rollback, audit trail and remote-support limits.
Fleet / Service Operations	Integrated dispatch, maintenance, charging, incident management, customer support and feedback from daily autonomous operations.	Connect the driving stack with fleet and service control; manage SLA, availability, intervention, energy and cost/hour.
Scale / Economics	Reusable software/hardware platforms, lower-cost sensors and compute, standardized integration and multi-vehicle deployment.	Track utilization, launch lead time, ODD rollout speed and lifecycle revenue; separate common and use-case-specific functions.

The industrial baseline combines a production-ready driver, closed-loop learning, scenario-based V&V, auditable safety, remote support, fleet operations and repeatable ODD deployment. The roadmap establishes these capabilities first, then differentiates through city data, infrastructure coordination, demand orchestration and business optimization.

AI-Controlled Mobility System

This section defines how AI functions as control logic across layers and is executed through autonomous driving systems. AI is applied differently at each layer — from demand prediction to system-wide coordination — but all functions are integrated through a unified control structure that enables real-time execution at city scale.

Layer	Cloud	Distributed Infrastructure	Fleet Vehicle
Perception	2030: Env-state estimation (fusion, uncertainty propagation) / 2035: City-wide world-state estimation (global sync, uncertainty minimization)	2030: Roadside perception augmentation (edge fusion, low-latency inference) / 2035: Cooperative perception (shared state, spatiotemporal fusion)	2030: Onboard perception (LiDAR / Camera / Radar, ≤10ms) / 2035: Infra-coordinated perception (shared scene state)
Planning	Dispatch optimization (forecast, constrained search) / 2035: City mission optimization (global scheduling, real-time recompute)	2030: Zone operation optimization (routing / berth / node) / 2035: Real-time replanning (node sync, global constraints)	2030: Local path / motion planning / 2035: Cooperative trajectory planning
Control	2030: Learning-based control (constraints, stability) / 2035: Global closed-loop optimization (adaptive)	2030: Edge feedback control (low-latency, constraints) / 2035: Distributed synchronized control (coordination, load)	2030: Motion control (MPC, ≤50ms, constraint-aware) / 2035: High-frequency adaptive control (≤10ms, cooperative)
Coordination	2030: Fleet-state coordination (dispatch-state sync) / 2035: Fleet-wide coordination (global sync, decision integration)	2030: Node-state coordination (local sync) / 2035: Node Mesh synchronization	2030: Vehicle-level coordination / 2035: Platoon coordination

ODD / DDT as Revenue Driver

ODD / DDT transition defines idle time and directly determines utilization and revenue structure.
DDT Loss = (1-TSR) x Recovery time + TOT, Revenue (OAR x UAC) x (1 ÷ (Trip+(1-TSR) x Recovery + TOT) x FLC x Load

Execution Sequence & Exit Gates

The roadmap advances only when technical evidence, operational readiness and unit economics are verified together.

Phase	Capability Build	Exit Gate	Business Outcome
1. Industrialize	Common driving and sensor stack; closed-loop data; scenario V&V; auditable safety case; remote operations and field support.	Repeatable launch within the initial ODD, with controlled fallback, rollback, incident response and configuration management.	Production service with measurable safety, availability, intervention rate, recovery time and cost/hour.
2. Federate	Fleet orchestration; RSU / OBU coordination; shared data and simulation; distributed infrastructure and service control.	Cross-vehicle and cross-site operation under common interfaces, authority boundaries, release procedures and KPI governance.	Lower intervention, recovery time and idle capacity; faster vehicle, site and ODD onboarding.
3. Scale	4D demand orchestration; Node Mesh; energy-compute integration; evidence-led quantum computing, sensing and optimization.	Multi-ODD, multi-service and multi-city performance with resilience, traceability and viable unit economics.	Scalable urban mobility platform with recurring service, infrastructure and lifecycle revenue.

Urban AI Operating Architecture

An integrated capability progression from real-world understanding and coordinated safety to governed learning and city-scale operations.

1. UNDERSTAND & PROTECT	2. CONNECT & IMPROVE	3. OPERATE & SCALE
Real-world intelligence and coordinated safety	Governed data, validation and continuous learning	Distributed execution and city-scale orchestration
AI Vision Engine — Builds a multimodal understanding of people, vehicles, infrastructure, road conditions and the surrounding urban environment.	Infra Hub — Connects roadside, mobility, building and public-service infrastructure through secure, governed and interoperable interfaces.	4D demand orchestration — Predicts where, when and why demand will emerge across mobility, logistics, energy and public services.
Behavior AI — Predicts human and vehicle intent, interaction patterns, anomalies and uncertainty to support context-aware decisions.	Data Fabric — Federates vehicle, city, infrastructure and operational data without forcing every participant into a single centralized system.	Rare-event learning — Prioritizes edge cases, near misses and low-frequency, high-impact events for targeted learning and validation.
Drive Sync Assist — Synchronizes driving support, shared awareness and timely intervention across the driver, vehicle and surrounding environment.	Digital KAIZEN — Converts operational evidence into simulation, validation, controlled deployment and measurable continuous improvement.	RSU / OBU coordination — Distributes perception, routing, safety and operational tasks across roadside and onboard systems.
Integrated ANZEN — Integrates detection, prediction, risk assessment and intervention, including graduated assistance, fallback and minimum-risk maneuvers.	Scenario simulation and validation — Tests policies, ODD changes, service patterns, traffic conditions and rare events before field deployment.	Node Mesh synchronization — Coordinates vehicles, roadside nodes and local service hubs as a resilient, distributed operating network.
Multimodal real-world understanding — Fuses cameras, sensors, maps, V2X signals and contextual data into a consistent operational picture.	Operational feedback loop — Captures outcomes, deviations, user behavior and system performance to update models, rules and operating procedures.	Energy-compute integration — Allocates charging, storage, edge computing and fleet tasks against demand, cost, capacity and resilience.
Predictive risk and uncertainty management — Identifies emerging conflicts early enough to preserve human decision time and maintain a safe operating margin.	MBD / digital-twin lifecycle loop — Links design assumptions, simulation results, field evidence and software or policy updates throughout the lifecycle.	Quantum computing and sensing — Explores optimization and sensing gains for high-dimensional mobility, infrastructure and urban-service operations.
Coordinated safety — Aligns human handover, vehicle fallback, infrastructure assistance and accountable decision-making across the operating environment.	Governed continuous improvement — Releases only validated updates with traceability, approval, rollback capability and measurable safety evidence.	Quantum-enhanced perception — Applies advanced sensing and inference only where they demonstrate meaningful gains over classical approaches.
Shared scene awareness — Makes relevant hazards, intent and operating context understandable across people, vehicles, infrastructure and service operators.	City-data integration — Connects mobility, energy, logistics and public-service operations through governed, reusable data products.	Full-city synchronization — Coordinates cross-domain actions while preserving local autonomy, safety boundaries and service continuity.
Safety-to-ODD expansion — Extends system capability only after evidence demonstrates acceptable risk and operational readiness in each target condition.	Closed-loop execution — Deploys approved models, policies and operating rules back to vehicles, RSUs, infrastructure and service operations.	Autonomous city-state optimization — Continuously balances safety, service quality, capacity, energy consumption, resilience and accessibility.
Human trust and adoption — Explains system state, preserves appropriate human authority and captures acceptance signals from users and operators.	Governance, KPI and business control — Monitors safety, availability, unit economics, energy, inclusion, service quality and lifecycle performance.	Scalable business execution — Converts technical gains into repeatable operations, partnerships, asset management and sustainable lifecycle value.

Cross-Layer Decision Rights & Release Gates

Control Domain	Primary Authority	Evidence Returned Upward	Release / Deployment Gate
Vehicle / Edge	Immediate perception, motion control, fallback, minimum-risk manoeuvre and local safety intervention.	Uncertainty, hazard traces, interventions, sensor and actuator health, and safe-state results.	Latency, hazard-response coverage, fail-operational capability and safe-state completion.
Fleet / Infrastructure	Dispatch, routing, RSU / OBU coordination, remote assistance, node control and energy-compute allocation.	Availability, dwell and recovery time, node congestion, service exceptions, energy use and capacity.	SLA, intervention rate, recovery time, operational continuity and ODD-support readiness.
Cloud / City	Model and policy release, scenario validation, governance, cross-domain learning and city-scale optimization.	Scenario coverage, causal evidence, audit trail, service quality, cost and lifecycle-performance KPIs.	Traceability, rollback readiness, validated KPI improvement, resilience and sustainable unit economics.

Validated policies, models and operating rules move downward; evidence, uncertainty, incidents and business KPIs move upward. Safety authority remains local, while learning and optimization remain globally governed, measurable and auditable. Across these three layers, the operating cycle remains Observe → Understand → Learn → Evaluate → Execute → Observe Again. Real-world signals are interpreted and protected at the edge; governed data is federated and converted into validated improvements; approved policies and models are then deployed back to vehicles, RSUs, infrastructure and urban operations. Evidence and learning move upward, while only validated policies and control updates move downward. The architecture scales from coordinated assistance to cross-domain orchestration across mobility, logistics, energy and public services, while preserving human accountability, fallback and rollback. Quantum computing, sensing and quantum-enhanced perception are introduced as evidence-led accelerators—not assumptions—and must demonstrate measurable gains in safety, resilience, service quality, capacity or lifecycle economics before deployment. The result is a practical path from isolated AI functions to a trusted, continuously improving urban operating system.

Satofumi Goto

Layer-integrated Control & Business Architecture

This architecture defines how business requirements are translated into executable system design through AI-driven control and autonomous driving. It connects KPI targets with technical execution, ensuring that system performance directly drives economic outcomes. Executable mobility operation This section maps business KPIs to control logic and AI-relevant technical conditions required to achieve them.

2030				2035		
	Logic	Input	Target	Logic	Input	Target
Perception	Match ≤500ms [A]	Cal ±5min	CVR ≥23%	Match ≤100ms [A]	Behavior ±1s	CVR ≥35%
	Update ≤5min	Res ≤1s	Wait ≤6min	Update ≤1s	History ≤10s	Wait ≤3min
	Trigger ≤1s	Avail ≤500ms	Cancel ≤12%	Trig ≤100ms	Avail ≤100ms	Cancel ≤5%
	Conv ≤1min		Eff conv≥19.6%	Conv ≤10s		Eff ≥33.3%
	Err ≤20%		Match ≤500ms	Err ≤5%		Matc ≤100ms
	Succ ≥85%		Err ≤20%	Reflect ≥95%		Err ≤5%
Planning			Trigger ≤1s			Trig ≤100ms
	Disp ≤1min [B]	Fleet ±50m	Trips ≥73/d	Disp ≤10s [B]	Fleet ±5m	Trips ≥140/d
	Route ≤5s	ETA err ≤3min	Loaded ≥14/d	Route ≤1s	Traffic ±1s	Load ≥112/d
	Alloc ≤10min	Traffic ≤30s	Resp ≤8min	Alloc ≤1min	Sync ≤5s	Resp ≤5min
	Loop ≤5min		ETA err ≤3min	Loop ≤1min		ETA err ≤2min
	Succ ≥95%		Empty ≤25%	E-int ≥95%		Empty ≤15%
Control			Util ≥75%	Succ ≥99%		Util ≥85%
			Route ≤5s			Route ≤1s
	MPC ≤50ms [C]	Percep ≥95%	Price ≥¥2,200	MPC ≤10ms [C]	Perce ≥99.9%	Price ≥¥2,000
	Traj ≤50ms	Plan ≤200ms	Ctrl ≤50ms	Traj ≤10ms	Plan ≤50ms	Ctrl ≤10ms
	Conv ≤100ms	V2X ≤20ms	Plan ≤200ms	Conv ≤10ms	Sync ≤10ms	Plan ≤50ms
	Stability ±15%		Err ≤30cm	Stability ±5%		Err ≤5cm
oordination	Fail-safe ≤100ms		Intv ≤1/100km	Coop ≥95%		Intv ≤1/10,000km
			Util ≥87.5%	Fail-safe ≤10ms		Util ≥95.8%
	Alloc ≤1min	Energy ±5% [D]	OPEX bound	Alloc ≤100ms	Energy ±1% [D]	OPEX≈¥202.8K/d
	Compute ≤1s	Comp ≤100ms	Comp ≤150ms	Comp ≤10ms	Comp ≤10ms	Comp ≤50ms
	Refuel ≤5min	Sync ≤1min	Eff ≥75%	Refuel ≤1min	Sync ≤10s	Eff ≥90%
	Conv ≤10min		Util ≥78%	Strategy ≤1h		Util ≥96%
	S-D ≥0.78		Eff energy ≥58.5%	Conv ≤1min		Eff energy ≥86.4%
	Load err ≤10%		Refuel ≤2/d	S-D ≥0.96		Refuel ≤1/d
			Var ≤±5%	Load err ≤3%		Var ≤±1%

Operating index (dimensionless) = Utilization × Load Factor × Energy Efficiency; profit still requires price, volume, and cost. Available service time = scheduled time – [(1–TSR)×recovery time + TOT]; Revenue = Trips × Seats × Load × Fare.

Layer-Integrated Business Architecture

This section defines business requirements and constraints., and translates them into executable system architecture. AI is applied as controlled logic across four layers.

2030			2035		
	Technology	KPI	Technology	KPI	
Service Orchestration	• Calendar-driven Demand Orchestration Engine	CVR ≥23%	• Behavioral AI Orchestration	CVR ≥35%	
	• Service Reservation	Wait ≤6min	• Agent-Demand Control	Wait ≤3min	
	• Availability Matching Engine	Cancel ≤12%	• 4D Digital Twin	Cancel ≤5%	
	• Reservation-triggered Dispatch	Match ≤500ms [A]	• Travel-time Monetization	Matches≤100ms [A]	
	• Linkage API	Avail ≤500ms	• City Service Optimization	Avail ≤100ms	
	• Customer Touchpoint PaaS	Res ≤1s	• Fleet Camera social sensing	Behavior ±1s	
	• Context-aware Notification	Cal ±5min	• IFTTT Trigger Dispatch	History ≤10s	
		Trigger ≤1s	• Autonomous Intent Agent	Reflect ≥98%	
Operation Control	• Dispatch Optimization (VRP)	Trips ≥73/d [B]	• Integrated TMS	Trips ≥140/d [B]	
	• Real-time Routing	Load ≥14/d	• Fleet-cooperative Control	Load ≥112/d	
	• Closed-loop Dispatch	Resp ≤8min	• Energy-Transport Optimizer	Resp ≤5min	
	• Demand Prediction	ETA err ≤3min	• Quantum-inspired Optimization	ETA err ≤2min	
	• Energy-aware Routing	Route ≤5s	• In / Out Energy Sync	Route ≤1s	
	• Demand-Supply Balancing	Alloc ≤10mi	• Real-time Replanning	Alloc ≤10s	
	• ETA Prediction	Fleet ±50m	• Constraint-aware Optimization	Fleet ±5m	
	• Capacity Allocation	Traffic ≤30s	• Multi-agent Coordination	Traffic ±1s	
	• ETA Feedback Control	Empty ≤25%	• Global State Synchronization	Empty ≤15%	
	• Queue Optimization	Util ≥75%	• Predictive Dispatch	Util ≥85%	
Fleet Vehicle	• Swappable Vehicle Module	Util ≥87.5% [C]	• Dynamic Vehicle Coupling	Util ≥95.8% [C]	
	• MS-SOFC + SSB	Err ≤30cm	• Onboard Mill+ AI Air Bat	Err ≤5cm	
	• LH2 Tank TypeIV Subframe	MPC ≤50ms	• HTS coil wound LH2 Tank	MPC ≤10ms	
	• MgH2 MH Main chassis	Traj ≤50ms	• H- Conductor adapted MH	Traj ≤10ms	
	• PMSM Distributed Drive	Conv ≤100ms	• Real-time energy optimization	Conv ≤10ms	
	• SiC Inverter	Stability ±15%	• Thermal-compute coupling	Stability ±5%	
	• SbW (Dual Motor Redundant)	Fail-safe ≤100ms	• High-frequency power control	Coop ≥98%	
	• BbW (Electro-Hydraulic +Regen)	Percep ≥95%	• Quantum-enhanced LiDAR	Fail-safe ≤10ms	
	• CFRP Leaf + Active Suspension	Plan ≤200ms	• Quantum magnetometer	Perce ≥99.9%	
	• CFRP Spoke + CTIS Tire	Intv ≤1/100km	• Atonic clock	Plan ≤50ms	
				Intv ≤1/10,000km	
Distributed Infrastructure	• High-precision docking	Eff ≥75%	• Precision docking	Eff ≥90%	
	• Wind-driven Mill	Util ≥78%	• Ultra-low latency V2X sync	Util ≥96%	
	• Wind-driven AMR	Energy ±5% [D]	• Water-pipeline Mill	Energy ±1% [D]	
	• SOFC + Ammonia Tank	Compute ≤1s	• Pipeline-integrated AMR	Compute ≤10ms	
	• Superconducting Power Cable	Sync ≤1min	• PCFC	Sync ≤1s	
	• 6G Communication Network	Refuel ≤5min	• H-conductive BIMH Storage	Refuel ≤1min	
	• Standardized energy interface	S-D ≥0.78	• HTS-coiled LH2 Pipeline	S-D ≥0.96	
	• Low-latency communication	Load err ≤10%	• Quantum Communication	Load err ≤3%	
	• Sub-ms time synchronization	Var ≤±5%	• Energy-transport interface	Var ≤±1%	

I design systems where business requirements and technical execution are structurally aligned.

System Design for Mobility Feasibility

This page defines scenario design targets for Energy Efficiency = 0.58 (=58%), Utilization = 0.75 (=18h/day ÷ 24h/day), and Load Factor = 0.70 (=2.8pax ÷ 4pax). They are not proven outcomes; each target requires metered validation through integrated system architecture.

The design is structured across energy supply [kWh/day], vehicle dynamics and response [ms, cm, Hz], and control and computation [ms, Hz]. Net traction demand is 2,080.5kWh/day (=28.5kWh/trip × 73trips/day). At 60% electrical efficiency, required fuel-LHV input is 3,467.5kWh/day and the hydrogen-equivalent requirement is 104.1kg/day (=3,467.5 ÷ 33.3). Annual PV energy can exceed annual net demand, but an annual ratio does not prove 24-hour supply without power, storage, loss, and weather modelling.

Vehicle and control design ensure "motion sickness free" operation, defined by control latency ≤20ms, decision latency ≤100ms, positioning error ≤10cm, and vibration ≤0.3Hz. These are maintained through superconductivity enabled by liquid hydrogen cooling and real-time control via HPC operating at 100Hz (=1 ÷ 0.01sec). The HPC value is a design target and does not by itself prove continuous satisfaction or ODD expansion.

Block	Variable	Corrected value	Formula / definition	KPI link
KPI	Energy Efficiency	0.58 target	$\Sigma \text{ useful traction output} \div \Sigma \text{ purchased energy input}$	Energy
	Utilization	0.75 (=18÷24)	18h/day ÷ 24h/day	Operation
	Load Factor	0.70 (=2.8÷4)	2.8pax ÷ 4pax	Demand
Constraint	Operation Hours	24h/day	operating-window target	Utilization
	Stability	0.95	1 – 0.05	Energy
	Throughput	40veh/h	60min ÷ 1.5min	Infra
Energy	Net traction demand	2,080.5kWh/day	28.5×73	Energy
	Fuel-LHV input	3,467.5kWh/day	2,080.5÷0.60	Energy
	SOFC electrical efficiency	0.60	net electricity ÷ fuel LHV	Energy
	H2 LHV	33.3kWh/kg	source constant	Energy
	H2-equivalent requirement	104.1kg/day	3,467.5÷33.3	Supply
	Annual net demand	759,383kWh/yr	2,080.5×365	Energy
	PV capacity	1,500kW	1.5MW×1,000	Energy
	PV yield	1,100kWh/kW/yr	scenario benchmark	Energy
	PV loss	0.15	15%	Energy
	PV generation	1,402,500kWh/yr	1,500×1,100×0.85	Energy
	Annual PV ratio	1.85	1,402,500÷759,382.5	Annual energy only
	24h supply gate	Not proven by annual ratio	hourly power + storage + loss + weather model	Infra
	Supply loss	0.05 target	5%	Energy
	Supply stability	0.95 target	supplied ÷ required energy	Energy
Node	Node Delay	≤50ms	measured end-to-end node delay	Infra
	Dispatch Interval	5sec	60sec ÷ 12	Utilization
	Trips	73/day target	scenario target; capacity ceiling 40×12=480/day	Demand
	Fleet	13 vehicles	ceil(73÷6)	Operation
	Load capacity	4pax	2×2 seats	Demand
	Energy per trip	28.5kWh/trip	2,080.5÷73	Energy
	Continuous ratio	1.00 target	24÷24	Utilization
	Availability	0.99 target	available time ÷ scheduled time	KPI
	Reliability	0.99 target	successful cycles ÷ total cycles	Stability
Vehicle	Control Sync	≤20ms	100ms ÷ 5	Motion sickness free
	Decision Sync	≤100ms	100ms	Motion sickness free
	Position Error	≤10cm	1m ÷ 10	Motion sickness free
	Platoon Stability	≤5%	0.05	Motion sickness free
	Control Cycle	100ms	1sec ÷ 10	Stability
	Response Delay	≤20ms	100ms ÷ 5	Motion sickness free
	Accel Jerk	≤0.5m/s ² /s	1 ÷ 2	Motion sickness free
	Vibration	≤0.3Hz	1Hz ÷ 3.33	Motion sickness free
Control	Sync Accuracy	0.99	1 – 0.01	Stability
	Prediction Rate	1.00 target	successful predictions ÷ attempts	Operation
	Planning Rate	1.00 target	successful plans ÷ attempts	Operation
	Execution Rate	100Hz	1 ÷ 0.01sec	Control
	HPC Latency	≤10ms	measured inference latency	Stability
	Node Sync	≤50ms	measured synchronization delay	Stability
UX	Flow Capacity	40veh/h	60÷1.5	Infra
	Utilization Gain	0.10 target	0.75–0.65	KPI
Comparator	PEMFC efficiency	~60% direct H2	DOE technology comparison; not 22%	No exclusion on efficiency alone
	Battery downtime	4h/day assumption	24–20	Scenario only
	Battery utilization	0.83 assumption	20÷24	Scenario only
	ICE efficiency	0.30 assumption	scenario benchmark	Scenario only
Physics	Annual energy ratio	1.85	PV annual generation ÷ annual net demand	Energy
	Continuous supply	Validation pending	hourly balance required	Utilization
	Conversion loss	0.40	1–0.60	Energy
System	Feasibility gate	Not proven by multiplication	all independent safety, energy, operation, and financial gates must pass	Final
Conclusion	Hydrogen equivalent	104.1kg/day	2,080.5÷0.60÷33.3	Core
	Node operation	Validation pending	24h hourly model required	Core
	Utilization	0.75 target	18÷24	Core
	KPI status	Validation pending	not 0.58×0.75×0.70	Final

Feasibility Conditions for Autonomous Mobility System

This page does not present financial outcomes. It defines scenario conditions that must be independently validated. Energy Efficiency = 0.58, Utilization = 0.75, and Load Factor = 0.70 are design targets, not derived facts. Feasibility depends on energy supply [kWh/day], available power [kW], storage duration [h], vehicle response [ms], control accuracy [cm], ODD evidence, and commercial constraints.

Powertrain alternatives must be compared on duty-cycle efficiency, degradation, refuelling/charging downtime, mass, packaging, infrastructure, and total cost. Direct-hydrogen PEM fuel cells are not excluded by a 22% efficiency assumption; DOE lists approximately 60% electrical efficiency for direct-H2 PEM. MS-SOFC remains the selected base-case candidate for this concept, subject to full TCO, durability, start-up, emissions, and ODD validation.

Node operation requires 2,080.5kWh/day of net traction energy. At 60% efficiency the fuel-LHV requirement is 3,467.5kWh/day, equivalent to 104.1kg-H2/day. PV generation of 1,402,500kWh/year gives an annual energy ratio of approximately 1.85, but continuous operation still requires an hourly supply-demand, storage, conversion-loss, and weather model.

"Motion sickness free" refers to a condition where control latency $\leq 20\text{ms}$, decision latency $\leq 100\text{ms}$, positioning error $\leq 10\text{cm}$, and vibration $\leq 0.3\text{Hz}$ are simultaneously satisfied. This condition determines user acceptance and directly enables Utilization = 0.75 ($=18\text{h/day} \div 24\text{h/day}$). HPC maintains control at an execution rate of 100Hz ($=1 \div 0.01\text{sec}$), ensuring continuous satisfaction of response constraints. The definition is retained as the design requirement; compliance must be demonstrated by test evidence.

Block	Variable	Value / status	Formula / validation	KPI link
KPI	Energy Efficiency	0.58 target	metered useful output $\div$ purchased input	Energy
	Utilization	0.75 target	$18 \div 24$	Operation
	Load Factor	0.70 target	$2.8 \div 4$	Demand
Constraint	Operation Hours	24h/day target	operating window	Utilization
	Stability	0.95 target	$1 - 0.05$	Energy
	Throughput	40veh/h	1 vehicle $\div$ 1.5min	Infra
Node	Net energy	2,080.5kWh/day	28.5×73	Energy
	Fuel input	3,467.5kWh/day	$2,080.5 \div 0.60$	Energy
	SOFC efficiency	0.60 target	net electricity $\div$ fuel LHV	Energy
	H2 LHV	33.3kWh/kg	constant	Energy
	H2 required	104.1kg/day	$3,467.5 \div 33.3$	Supply
	Annual net energy	759,383kWh/yr	$2,080.5 \times 365$	Energy
	PV capacity	1,500kW	$1.5\text{MW} \times 1,000$	Energy
	PV yield	1,100kWh/kW/yr	benchmark assumption	Energy
	PV loss	0.15	15%	Energy
	PV generation	1,402,500kWh/yr	$1,500 \times 1,100 \times 0.85$	Energy
	Annual PV ratio	1.85	$1,402,500 \div 759,382.5$	Annual only
	Continuous supply	Not yet demonstrated	hourly balance + storage + reserve	Utilization
	Supply stability	0.95 target	supplied $\div$ required	Energy
	Node delay	$\leq 50\text{ms}$ target	measured	Infra
	Dispatch interval	5sec	$60 \div 12$	Utilization
	Trips	73/day target	scenario target	Demand
	Fleet	13	$\text{ceil}(73 \div 6)$	Operation
	Load	4pax	2×2	Demand
	Energy/trip	28.5kWh	$2,080.5 \div 73$	Energy
	Reliability	0.99 target	successful cycles $\div$ total	Stability
	Availability	0.99 target	available $\div$ scheduled time	KPI
Vehicle	Control Sync	$\leq 20\text{ms}$	$100\text{ms} \div 5$	Motion sickness free
	Decision Sync	$\leq 100\text{ms}$	100ms	Motion sickness free
	Position Error	$\leq 10\text{cm}$	$1\text{m} \div 10$	Motion sickness free
	Platoon Stability	$\leq 5\%$	0.05	Motion sickness free
	Control Cycle	100ms	$1\text{sec} \div 10$	Stability
	Response	$\leq 20\text{ms}$	$100\text{ms} \div 5$	Motion sickness free
	Jerk	$\leq 0.5\text{m/s}^2/\text{s}$	$1 \div 2$	Motion sickness free
	Vibration	$\leq 0.3\text{Hz}$	$1\text{Hz} \div 3.33$	Motion sickness free
	Sync Accuracy	0.99	$1 - 0.01$	Stability
Control	Prediction	1.00 target	successful prediction ratio	Operation
	Planning	1.00 target	successful plan ratio	Operation
	Execution	100Hz	$1 \div 0.01\text{sec}$	Control
	HPC latency	$\leq 10\text{ms}$ target	measured	Stability
	Node sync	$\leq 50\text{ms}$ target	measured	Stability
	Flow	40veh/h	$60 \div 1.5$	Infra
UX	Utilization Gain	0.10 target	0.75–0.65	KPI

This defines the conditions under which autonomous mobility may become operationally feasible. Passing one ratio does not substitute for validation of the remaining gates.

Proposed Responsibility Structure for Autonomous Mobility System

This is a proposed RACI hypothesis for discussion, not a statement of current organizational accountability. Final responsibility must be confirmed by corporate governance, contracts, permits, product-liability allocation, operational rules, and the approved safety case. Continuous operation depends on explicit ownership across orchestration, fleet operation, vehicle/system safety, node execution, energy supply, commercial settlement, and finance. A method such as model-based development is not an accountable legal entity; each task must be assigned to an organization or contracted operator.

Domain	Function / task	Accountable (A)	Responsible (R)	Support / consulted	Status
Demand	Generate demand	Mobility Platform	Mobility Platform	AD Platform / partners	Proposed
	Capture demand	Mobility Platform	Mobility Platform	MaaS Operator	Proposed
	Set pricing	Commercial owner	Commercial owner	Finance / regulator	Contract required
	Approve pricing	Vehicle OEM / business owner	Commercial owner	Legal / regulator	Approval required
	Optimize load	MaaS Operator	Mobility Platform	AD Platform	Proposed
	Manage UX	Mobility Platform	Mobility Platform	AD Platform / users	Proposed
	Define service	Vehicle OEM / business owner	Mobility Platform	MaaS Operator / user	Proposed
Operation	Execute dispatch	MaaS Operator	MaaS Operator	Mobility / AD Platforms	Permit + SLA
	Operate fleet	MaaS Operator	MaaS Operator	OEM / AD Platform	Permit + SLA
	Maintain service	MaaS Operator	MaaS Operator	Node operator / Mobility Platform	SLA
	Manage utilization	MaaS Operator	Mobility Platform	AD Platform	Proposed
	Define hours	MaaS Operator	MaaS Operator	Regulator / node operator	Permit
	Control dispatch	MaaS Operator	AD control platform	Mobility Platform	Authority boundary
Vehicle	Define architecture	Vehicle OEM	Vehicle OEM	AD Platform / Tier-1	Product governance
	Supply platform	Vehicle OEM	Vehicle OEM	Tier-1 suppliers	Commercial contract
	Supply drivetrain	Vehicle OEM	Tier-1 supplier(s)	AD Platform	Commercial contract
	Package cabin	Vehicle OEM	Body integrator / supplier	Mobility Platform	Commercial contract
	Maintain vehicle	Fleet owner / MaaS Operator	Certified service provider	Vehicle OEM	SLA
	Realize control	Vehicle OEM	AD Platform + Tier-1	MaaS Operator	Safety case
	Maintain position	AD Platform / OEM	AD Platform	Mobility Platform	Safety case
Facility	Provide site	Node / Parking Operator	Node / Parking Operator	Vehicle OEM	Contract
	Operate facility	Node / Parking Operator	Node / Parking Operator	MaaS Operator	Permit + SLA
	Sell services	Commercial owner	Mobility Platform / provider	Node operator	Contract
	Use platform	MaaS Operator	MaaS Operator	AD Platform	SLA
	Manage circulation	Node operator	Node operator	Vehicle OEM	SLA
Service	Provide booking	Mobility Platform	Mobility Platform	AD Platform	Service governance
	Integrate service	Mobility Platform	Mobility Platform	Service providers	API contract
	Sell package	Commercial owner	Commercial owner	Finance	Contract
	Optimize stays	MaaS Operator	Mobility Platform	Service providers	Proposed
	Operate service	Service provider	Service provider	MaaS Operator	SLA
MaaS	Operate platform	MaaS Operator	AD + Mobility Platforms	Vehicle OEM	Authority boundary
	Use platform	MaaS Operator	MaaS Operator	AD Platform	SLA
External	Provide entry	External platform owner	External platform owner	Mobility Platform	API contract
	Send traffic	External platform owner	External platform owner	MaaS Operator	Commercial contract
	Receive traffic	MaaS Operator	Mobility Platform	External platform	Commercial contract
Energy	Define supply	Energy business owner	Energy integrator / supplier	AD Platform / node operator	Contract
	Produce hydrogen	Energy supplier	Energy supplier	Node operator	Permit + quality
	Distribute energy	Energy supplier	Energy integrator / supplier	Node operator	Contract
	Manage logistics	Energy Integrator	Energy Integrator	Energy supplier	Contract
	Operate energy	Node operator	Node operator	AD Platform	SLA
Software	Define scope	Vehicle OEM / product owner	AD Platform	Mobility Platform	Product governance
	Develop control	AD Platform	AD Platform	Vehicle OEM / Tier-1	Release governance
	Operate compute	AD Platform	Cloud / edge operator	MaaS Operator	SLA
	Maintain dispatch	MaaS Operator	AD + Mobility Platforms	Node operator	SLA
Finance	Collect mobility	Commercial owner	Finance / payment operator	Mobility Platform	Settlement contract
	Collect facility	Node operator	Finance / payment operator	Mobility Platform	Settlement contract
	Collect energy	Energy business owner	Finance / payment operator	Energy Integrator	Settlement contract
	Maintain projection	Business owner	Finance	Vehicle OEM	Governance
	Collect data	Data controller	Mobility Platform	AD Platform / operators	Privacy + data contract
	Collect facility data	Node operator	Mobility Platform	AD Platform	Privacy + data contract

Responsibility is complete only after the accountable legal entity, operational authority, safety authority, data controller, and revenue/cost owner are all confirmed.

Pricing Structure and Billing Rules for Autonomous Mobility System

The amounts below remain illustrative scenario tariffs, not verified market prices. To prevent double counting, one base trip fare is selected per trip; add-ons apply only when the corresponding service is consumed; subscriptions offset eligible base charges; B2B platform fees are charged only when not already embedded in another contract; dividends/revenue shares are settlements below revenue, not customer prices.

Revenue, not customer price.							
Category	Payer	Payee	Charge	Billing rule	Illustrative formula / annualization	PL mapping	
User	User	Service provider	Facility Fee	Add-on	¥500/visit×200/day×365 = ¥36.500M/yr	MaaS service	
			Premium Service	Add-on	¥1,500/booking×120/day×365 = ¥65.700M/yr	Service revenue	
			Add-on	Add-on	¥600/use×200/day×365 = ¥43.800M/yr	Service revenue	
	MaaS Operator	MaaS Operator	Service Fee	Base fare (exclusive)	¥1,000/trip×130/day×365 = ¥47.450M/yr	Trip revenue	
			Trip Fee	Base fare (exclusive)	¥2,000/trip×100/day×365 = ¥73.000M/yr	Trip revenue	
			Premium Trip	Base fare (exclusive)	¥2,500/trip×90/day×365 = ¥82.125M/yr	Trip revenue	
			Subscription	Offsets eligible base fares	¥15,000/user×1,200 users = ¥18.000M/yr	Subscription revenue	
			Night Trip	Base fare (exclusive)	¥2,700/trip×80/day×365 = ¥78.840M/yr	Trip revenue	
	Parking Operator	Parking Operator	Premium Facility	Add-on	¥800/visit×120/day×365 = ¥35.040M/yr	Facility revenue	
			Loyalty Fee	Add-on / retention	¥200/visit×185/day×365 = ¥13.505M/yr	Service revenue	
Facility	AD Platform	Parking Operator	Site Rent	B2B fixed fee	¥300,000/month×30×12 = ¥108.000M/yr	Node OPEX	
			Operating Access	B2B access fee	¥700/visit×90/day×365 = ¥22.995M/yr	Node OPEX	
	Parking Operator	Service Fee	B2B service	¥300/visit×200/day×365 = ¥21.900M/yr	Node revenue		
	Service Provider	Parking Operator	Upgrade Rent	B2B fixed fee	¥2,000,000/month×30×12 = ¥720.000M/yr	Node OPEX	
Service	Service Provider	Mobility Platform	Booking Fee	Per booking	¥100/booking×150/day×365 = ¥5.475M/yr	Reservation fee	
			CRM Fee	B2B fixed fee	¥80,000/month×30×12 = ¥28.800M/yr	Platform fee	
			API Fee	B2B usage fee	¥80/call×250/day×365 = ¥7.300M/yr	Platform fee	
			Premium UX	B2B product fee	¥250/booking×180/day×365 = ¥16.425M/yr	Platform fee	
			AD Platform	Optimization Fee	B2B product fee	¥50/call×300 calls/day×365 = ¥5.475M/yr	Platform fee
			Premium API	B2B usage fee	¥150/call×60/day×365 = ¥3.285M/yr	Platform fee	
			Sync Fee	B2B product fee	¥12/call×5,500/day×365 = ¥24.090M/yr	Platform fee	
			Service Provider	Late Fee	Customer charge	¥1,200/use×70/day×365 = ¥30.660M/yr	Service revenue
	Mobility	MaaS Operator	AD Platform	Control Fee	B2B product fee	¥100/dispatch×100/day×365 = ¥3.650M/yr	Control OS
				Routing Fee	B2B product fee	¥120/dispatch×110/day×365 = ¥4.818M/yr	Platform fee
Mobility Platform		UX Fee	B2B product fee	¥50/trip×100/day×365 = ¥1.825M/yr	Platform fee		
		Parking Operator	Advanced Control	B2B service fee	¥150/dispatch×90/day×365 = ¥4.928M/yr	Node service	
AD Platform		MaaS Operator	Incentive	Contra-revenue / settlement	¥200,000/month×30×12 = ¥72.000M/yr	Settlement; not gross revenue	
User		MaaS Operator	Surge Pricing	Base fare adjustment	¥2,800/trip×80/day×365 = ¥81.760M/yr	Trip revenue	
	Off-peak Discount		Base fare adjustment	¥1,600/trip×70/day×365 = ¥40.880M/yr	Trip revenue		
Platform	MaaS Operator	AD Platform	OS Fee	B2B fixed fee	¥80,000/month×30×12 = ¥28.800M/yr	Platform fee	
			Monthly OS	B2B fixed fee	¥250,000/month×30×12 = ¥90.000M/yr	Platform fee	
	Service Provider	AD Platform	Node Fee	B2B fixed fee	¥200,000/month×30×12 = ¥72.000M/yr	Node service	
			Data Fee	B2B usage fee	¥100,000/month×30×12 = ¥36.000M/yr	Platform fee	
	Parking Operator	AD Platform	Facility Fee	B2B fixed fee	¥200,000/month×30×12 = ¥72.000M/yr	Node service	
			Analytics Fee	B2B fixed fee	¥120,000/month×30×12 = ¥43.200M/yr	Platform fee	
			Control OS	B2B product fee	¥220/trip×110/day×365 = ¥8.833M/yr	Control OS	
			Booking Link	B2B usage fee	¥180,000/month×30×12 = ¥64.800M/yr	Reservation fee	
External	External Platform	AD Platform	API Fee	B2B usage fee	¥120/use×5,000/day×365 = ¥219.000M/yr	Platform fee	
			IoT Fee	B2B usage fee	¥90/booking×200/day×365 = ¥6.570M/yr	Platform fee	
			Tx Fee	B2B usage fee	¥200/booking×100/day×365 = ¥7.300M/yr	Platform fee	
			Premium API	B2B usage fee	¥150/call×6,000/day×365 = ¥328.500M/yr	Platform fee	
			Premium Tx	B2B usage fee	¥250/booking×180/day×365 = ¥16.425M/yr	Platform fee	
			Premium Referral	Revenue share / settlement	15% of eligible booking revenue	Settlement; not gross revenue	
			Sync Fee	B2B usage fee	¥12/call×5,500/day×365 = ¥24.090M/yr	Platform fee	
Energy	AD Platform	Finance Co.	Energy Fee	B2B service	20kWh/trip×100/day×¥30×365 = ¥21.900M/yr	Energy revenue	
			Peak Energy	B2B service	25kWh/trip×90/day×¥45×365 = ¥36.956M/yr	Energy revenue	
			Smart Energy	B2B service	¥70/use×120/day×365 = ¥3.066M/yr	Energy revenue	
	Service Provider	AD Platform	Energy Opt	B2B product fee	¥50/use×150/day×365 = ¥2.738M/yr	Platform fee	
Finance	MaaS Operator	Finance Co.	Base Energy	B2B settlement	18kWh/trip×110/day×¥28×365 = ¥20.236M/yr	Energy OPEX	
			Lease	B2B lease	¥3,000,000/year×365-day basis = contract specific	Lease / CAPEX	
			Dividend	Settlement only	profit distribution after tax	Not revenue	
			Node Lease	B2B lease	¥4,500,000/year per node	Lease / CAPEX	
			Node Lease	B2B lease	¥2,500,000/year per node	Lease / CAPEX	
Finance	Service Provider	Finance Co.	Lease	B2B lease	¥2,000,000/year×365-day basis = contract specific	Lease / CAPEX	
			Dividend	Settlement only	profit distribution after tax	Not revenue	
			Service Lease	B2B lease	¥2,500,000/year per asset	Lease / CAPEX	

Only lines explicitly mapped to P15 may enter the PL. Settlement lines and mutually exclusive base fares must not be added again.

Autonomous Mobility Business Control Structure

System conditions define the feasibility of control and set operational boundaries. Values below are phase targets or scenario assumptions; they are not externally verified outcomes. Minimum gates on P1 and phase targets here are different by design and must be labelled separately.

Revenue is calculated with unit-consistent formulas. When fare is per passenger, annual revenue = Trips/day × Seats × Load factor × Fare/passenger × 365. Break-even trips/day cannot be calculated without a defined annual controllable-cost allocation.

Layer	Control item	Decision logic	2030	2035	PL impact
Demand	Demand generation	Requests ÷ potential demand	≥0.50	≥0.70	Trips
	Conversion	Bookings ÷ requests	≥0.70	≥0.85	Trips
	Matching	Matched ÷ requests	≥0.85	≥0.95	Trips
	Trip realization	Executed ÷ matched	≥0.80	≥0.95	Trips
	Stabilization	Matched trips + planned demand	≥0.80	≥0.95	Trips
	Time synchronization	Calendar-based demand	≥0.60	≥0.85	Trips
	Service capture	Reservation-based trips ÷ total	≥0.50	≥0.75	Trips
	Demand-cap control	Demand ≤ system capacity	100%	100%	Trips cap
Operation	Trips	Executed trips/day	73	140	Trips
	Utilization	Operating time ÷ 24h	≥0.75	≥0.85	Trips
	Dispatch success	Successful dispatch ÷ requests	≥0.95	≥0.99	Trips
	Supply-body cycle	Active body ÷ total cycle	≥0.80	≥0.95	Utilization
	Waiting time	Passenger request-to-pickup	≤360sec	≤180sec	Trips
	Turnaround	Idle + service time	≤25min	≤10min	Utilization
	Queue control	Total wait ÷ requests	≤300sec	≤120sec	Conversion
	Optimization	Served ÷ queued	≥0.90	≥0.98	Trips
	Scheduled continuity	Continuous-operation ratio	≥0.80	≥0.95	Utilization
	Operation cap	Operation ≤24h	Fixed	Fixed	Cap
Vehicle	Latency control	Control delay	≤100ms	≤50ms	Dispatch
	Stability	Deviation ÷ target	≤15%	≤5%	Safety
	Intervention	Interventions ÷ km	≤0.010/km	≤0.0001/km	Utilization
	Motion response	Control latency / planning / ride targets	Design gate	Design gate	Utilization
	Response accuracy	Correct ÷ total control	≥0.90	≥0.98	Dispatch
	Platoon composition	Vehicles per formation	≥1.5	≥3.0	Load
	Ride continuity	Continuous ride + trips	≥0.85	≥0.90	Utilization
	Thermal constraint	Temperature limit	Fixed	Improved	Operation
	Vehicle availability	Available fleet ÷ total	≥0.80	≥0.95	Trips
Energy	End-to-end efficiency	Useful output ÷ purchased input	0.58 target	0.65 target	Cost
	Energy cost/trip	Cost ÷ trips	≤¥500	≤¥350	Cost
	Supply stability	Continuous supplied ÷ required	≥0.95	≥0.99	Utilization
	H2-equivalent requirement	Net energy ÷ efficiency ÷33.3	104.1kg/day	171.2kg/day	Trips cap
	PV role	Annual energy contribution	Annual ratio only	Annual ratio only	Cost
	Storage capacity	Hourly storage / reserve	Required	Required	Trips cap
	Distribution	Loss + supply	≤10%	≤5%	Cost
	Node allocation	Allocated ÷ demand	≥0.90	≥0.98	Trips
	Energy scheduling	Scheduled + required	≥0.85	≥0.95	Trips
System	Trip generation	Demand×Matching×Dispatch×Utilization	73/day	140/day	Revenue
	Load optimization	Passengers ÷ capacity	0.70	0.80	Revenue
	Fare	JPY per passenger	¥800	¥900	Revenue
	Throughput	Trips ÷ hour	≥1.5	≥3.0	Revenue
	Density	Trips ÷ area	Low	High	Revenue
	Scaling factor	Trips 2035 ÷ 2030	1.00×	1.92×	Revenue
	Energy linkage	28.5kWh × trips	2,080.5kWh	3,990kWh	Cost
	Hydrogen linkage	Energy ÷ efficiency ÷33.3	104.1kg	171.2kg	Cost
	Fleet sizing	ceil(trips ÷6)	13	24	Trips cap
	Congestion	Load ÷ capacity	Medium	High	Limit
	Node capacity	Throughput ≤ capacity	Limited	Expanded	Cap
Financial	Annual passenger revenue	Trips×Seats×Load×Fare×365	¥59.685M	¥183.960M	Profit
	Revenue per vehicle trip	Seats×Load×Fare	¥2,240	¥3,600	Revenue
	Contribution	Allocated revenue – allocated cost	See P15	See P15	IRR
	Break-even trips/day	Annual controllable cost ÷ (Seats×Load×Fare×365)	TBD	TBD	Profit
	Cost control	Energy + operation + vehicle + infrastructure	High	Reduced target	Cost
	Price constraint	Applicable fare / contract rules	Validate	Validate	Revenue
	IRR hurdle	Return target	≥10%	≥12%	Investment

This defines unit-consistent control logic connecting operational conditions to trips, revenue, cost, and investment gates.

Autonomous Mobility Business Profit & Loss Structure (2030 / 2035)

This is a scenario PL, not a forecast. Each monetary line is recalculated from the displayed formula. Non-monetary operating drivers are separated from revenue. The prior ¥36B node-sale line is excluded from the base case because asset-transfer price, delivery cost, and COGS were not defined; recurring node-operation service remains included.

Layer	Category	Item	2030 formula	2030	2035 formula	2035		
Revenue	Energy	Elec Internal	4.0M×20%×50%×1,200×¥15	¥7.200B	4.0M×30%×50%×1,500×¥15	¥13.500B		
		Elec External	4.0M×25%×50%×1,200×¥15	¥9.000B	4.0M×35%×50%×1,500×¥15	¥15.750B		
		H2 Internal	4.0M×80%×1%×67×¥333	¥0.714B	4.0M×70%×20%×67×¥333	¥12.494B		
		H2 External	4.0M×75%×1%×67×¥333	¥0.669B	4.0M×65%×20%×67×¥333	¥11.602B		
		Titanium	80,000×0.3×¥72,000	¥1.728B	same	¥1.728B		
		Node hardware sale	Excluded: price and COGS undefined	¥0.000B	Excluded: price and COGS undefined	¥0.000B		
		Driver	24H Utilization	18÷24	0.75	22÷24	0.92	
Load Factor	2.8÷4		0.70	4.5÷5	0.90			
Swap-body Util.	Active÷Total		0.80	Dynamic composition	0.95			
Night Operation	Night revenue÷Total		0.35	Continuous operation	0.60			
Auto Loading Rev.	Loading fee rev.÷Total		0.22	Integrated logistics	0.35			
Revenue	MaaS		Night Passenger	25×24×¥15,000×365×60%	¥1.971B	500×24×¥15,000×365×60%	¥39.420B	
		Night Cargo	25×24×¥5,000×365×40%	¥0.438B	500×24×¥5,000×365×40%	¥8.760B		
		Mobile Storage	400×¥40,000×12	¥0.192B	5,000×¥40,000×12	¥2.400B		
Platform	Auto Loading	269,000×¥22,300	¥5.999B	538,000×¥22,300	¥11.997B			
	Platform Fee	Dispatch/API fee	¥0.800B	Integrated OS fee	¥6.500B			
	Reservation Fee	Booking linkage	¥0.350B	Integrated booking	¥2.200B			
	Node Service	Node operation fee	¥0.420B	Node ecosystem service	¥4.800B			
Total	Total Revenue	sum monetary revenue	¥29.481B	sum monetary revenue	¥131.151B			
OPEX	Energy	H2 Internal COGS	4.0M×80%×1%×67×¥150	▲¥0.322B	4.0M×70%×20%×67×¥150	▲¥5.628B		
		H2 External COGS	4.0M×75%×1%×67×¥150	▲¥0.302B	4.0M×65%×20%×67×¥150	▲¥5.226B		
		Elec Internal	4.0M×20%×50%×1,200×¥6.7	▲¥3.216B	4.0M×30%×50%×1,500×¥6.7	▲¥6.030B		
		Elec External	4.0M×25%×50%×1,200×¥6.7	▲¥4.020B	4.0M×35%×50%×1,500×¥6.7	▲¥7.035B		
		Titanium	80,000×0.3×¥6,000	▲¥0.144B	same	▲¥0.144B		
		Material Cycle	scenario cost	▲¥0.300B	onboard circulation	▲¥1.800B		
		Node Optimization	distributed control	▲¥0.120B	integrated mesh	▲¥0.950B		
		Energy Sync	In/Out sync	▲¥0.080B	real-time optimization	▲¥0.600B		
MaaS	MaaS Ops	25×¥81.0M	▲¥2.025B	500×¥45.6M	▲¥22.800B			
	Dispatch Ops	fleet dispatch	▲¥0.550B	integrated dispatch	▲¥4.200B			
	Remote Ops	remote support	▲¥0.300B	integrated operation	▲¥2.100B			
Platform	Platform Ops	¥1.0B/year	▲¥1.000B	¥6.0B/year	▲¥6.000B			
	Cloud Ops	cloud/HPC	▲¥0.500B	distributed HPC cloud	▲¥3.800B			
	AI Ops	AI orchestration	▲¥0.220B	integrated AI control	▲¥2.400B			
Total	Total OPEX	sum	▲¥13.098B	sum	▲¥68.713B			
SG&A	Energy	Energy SG&A	energy product revenue×10%	▲¥1.931B	energy product revenue×10%	▲¥5.507B		
		Maintenance	scenario	▲¥0.250B	integrated maintenance	▲¥1.800B		
MaaS	MaaS SG&A	MaaS revenue×15%	▲¥0.390B	MaaS revenue×15%	▲¥7.587B			
	Customer Ops	user support	▲¥0.180B	integrated support	▲¥1.500B			
Platform	Platform SG&A	platform revenue×20%	▲¥1.514B	platform revenue×20%	▲¥5.099B			
	Partner Ops	alliance operation	▲¥0.120B	integrated alliance	▲¥1.100B			
Total	Total SG&A	sum	▲¥4.385B	sum	▲¥22.594B			
Profit	Total	EBITDA	Revenue–OPEX–SG&A	¥11.998B	Revenue–OPEX–SG&A	¥39.844B		
Dep.	Energy/MaaS/Platform	Node	20.000B÷15y	▲¥1.333B	180.000B÷15y	▲¥12.000B		
		GTFC	18.450B÷15y	▲¥1.230B	33.450B÷15y	▲¥2.230B		
		Mill	0.600B÷10y	▲¥0.060B	15.600B÷10y	▲¥1.560B		
		Battery	5.000B÷10y	▲¥0.500B	10.000B÷10y	▲¥1.000B		
		Magnetic	3.000B÷10y	▲¥0.300B	6.000B÷10y	▲¥0.600B		
		HTS	4.000B÷10y	▲¥0.400B	8.000B÷10y	▲¥0.800B		
		Water	2.000B÷10y	▲¥0.200B	4.000B÷10y	▲¥0.400B		
		HPC	5.000B÷15y	▲¥0.333B	45.000B÷15y	▲¥3.000B		
		Energy Device	Mill/Batt/HTS group	▲¥0.600B	Integrated energy device	▲¥3.600B		
		Vehicle Dev	50.000B÷5y	▲¥10.000B	100.000B÷5y	▲¥20.000B		
		Prototype	10.000B÷5y	▲¥2.000B	10.000B÷5y	▲¥2.000B		
		Swap-body Dev	Swap-body architecture	▲¥1.500B	Dynamic coupling	▲¥5.200B		
		Platform Dev	40.000B÷5y	▲¥8.000B	110.000B÷5y	▲¥22.000B		
		AI Platform	AI orchestration	▲¥1.800B	Integrated AI infra	▲¥8.500B		
		Control OS	Dispatch/control OS	▲¥1.200B	City OS integration	▲¥6.200B		
		Infra Sync	Node synchronization	▲¥0.450B	Integrated node mesh	▲¥2.600B		
		Revenue Logic	24H utilization logic	▲¥0.300B	Dynamic load logic	▲¥1.500B		
		Total	Total Depreciation	sum	▲¥30.206B	sum	▲¥93.190B	
		Profit	Total	Operating Profit	EBITDA–Dep.	▲¥18.208B	EBITDA–Dep.	▲¥53.346B
				Contribution before depreciation	Revenue–OPEX–SG&A	¥11.998B	Revenue–OPEX–SG&A	¥39.844B
Operating Margin	Operating Profit÷Revenue			-61.8%	Operating Profit÷Revenue	-40.7%		
Project Payback	see P18			2035 nominal	see P18	discounted not achieved by 2035		
Project IRR	reconciled cash flow			0.47%	12% hurdle	not met		

Assumptions retained unless corrected above: city 1/5; MaaS vehicles 25/500; nodes 4/36; passenger cars 4.0M/year; trucks 80,000/year; electricity sales ¥15/kWh and cost ¥6.7/kWh; H2 sales ¥333/kg and cost ¥150/kg; titanium exchange 80,000×0.3, sales ¥72,000 and cost ¥6,000. These are scenario assumptions requiring primary-source validation.

2030 Vehicle Platform

A 12m autonomous MaaS skateboard uses an MS-SOFC scenario with city gas 70% and mill-derived hydrogen 30%. Hydrogen is supplied from stationary/node-side milling; batteries support transient power. Component ranges sum to approximately 9.8–15.1t before the stated 10–15% integration reserve; the upper case approaches 17.3t. Therefore, payload is not claimed until body, fluids, passengers, wiring/piping, axle loads, and certified curb mass are included.
The 10–15kg LH2 buffer requires approximately 340L after the 1.6 packaging factor. The envelope is corrected to 450×550×1,400mm (346.5L external envelope). For 240kW electrical output at 60% efficiency, fuel input is 400kW and nominal rejected heat is approximately 160kW, not 96kW.

Component	Qty	Performance	Formula	Corrected result	H/W/D mm	Weight	Layout
1 Energy/H2/Power		12m flat skateboard	-	upper container unused	-	-	overall
Total energy demand	1	1.2kWh/km	$1.2 \times 1,000$	1,200kWh	-	-	overall
Required fuel energy	1	MS-SOFC 60%	$1,200 \div 0.60$	2,000kWh-LHV	-	-	overall
Hydrogen ratio	1	H2 30% blend	$2,000 \times 0.30$	600kWh-H2	-	-	overall
Required H2	1	33.3kWh/kg	$600 \div 33.3$	18.0kg-H2	-	-	overall
City-gas ratio	1	70%	$2,000 \times 0.70$	1,400kWh-gas	-	-	overall
LH2 buffer	1	10–15kg	reserve	10–15kg-H2	450/550/1400	150–250kg	left
LH2 installed volume	1	insulation×1.6	$15 \div 70.8 \times 1.6$	≈340L	450/550/1400	included	left
City-gas tank	2	blended-fuel support	-	operation reserve	300/400/1600	300–500kg	left
Water tank	2	cooling/process water	-	≈180L	250/400/1200	180kg	left
Process-water control	2	steam-to-carbon / condensate	validated setpoint	not NOx claim	180/250/400	80–120kg	front
O2 membrane	2	optional O2 management	test gate	efficiency TBD	180/250/450	80–120kg	front
CO2 membrane	2	reformat cleanup	70–90% target	CO2 removal target	220/250/500	100–150kg	rear
Steam membrane	2	H2O separation	test gate	reforming support	180/250/450	80–120kg	rear
Reformer	2	steam reforming	fuel-path model	H2+CO reformat	260/350/700	250–350kg	front
Gas/liquid separator	2	50L/min	50×2	100L/min	220/250/450	120–180kg	F/R
H2 purifier	2	50L/min	50×2	100L/min	220/250/500	120–180kg	F/R
Vaporizer	2	20kg/h	20×2	40kg/h	240/300/600	120–180kg	left
LH2 pump	2	50L/min	50×2	100L/min	180/220/350	60–100kg	left
Pressure regulator	4	high→low P	-	4 systems	120/200/300	40–60kg	overall
Shutoff valve	6	safety shutoff	-	6 systems	120/180/250	40–60kg	overall
Vent line	2	safety vent	-	2 systems	100/100/1400	20–40kg	upper
Safety sensors	1	H2/P/T monitor	-	integrated	80/200/300	<30kg	overall
2 Electric/Thermal		MS-SOFC power control	-	battery transient	-	-	center
MS-SOFC modules	6	40kW/unit	40×6	240kW	320/1000/1400	1,200–1,800kg	center
MS-SOFC efficiency	1	electrical efficiency	-	≈60% target	-	-	center
Inverter	2	120kW	120×2	240kW	180/420/650	180–260kg	upper center
DC/DC	2	100kW	100×2	200kW	160/350/500	120–180kg	upper center
Solid-state battery	1	400kWh	-	400kWh	260/1200/2200	2,500–3,500kg	center
HV J-BOX	1	HV distribution	-	1 system	180/350/500	40–80kg	center
EMS/VCU	1	energy+vehicle ctrl	-	1 system	120/250/300	40–80kg	center
Radiator	2	rejected heat	$240 \times (1/0.60 - 1)$	≈160kW	180/500/1400	180–300kg	front
Cooling pump	2	50L/min	50×2	100L/min	160/180/250	40–80kg	overall
Waste-heat recovery	1	Rankine/thermal	validated recovery	aux. power TBD	260/600/900	250–450kg	rear
Insulation monitoring	1	HV safety	-	1 system	80/200/300	<20kg	center
Power protection	1	fuse/breaker	-	integrated	80/200/300	<20kg	center
3 AD Control		autonomous execution	-	AD operation	-	-	upper
AD computer	1	prediction/planning	latency≤100ms	control target	120/400/500	40–80kg	upper
Sensor suite	1	LiDAR/camera/radar	-	360° sensing	120/500/800	40–80kg	upper
Fusion/localization	1	map+IMU+GNSS	-	position control	80/300/400	<30kg	upper
ODD/DDT monitor	1	state-dependent safety gate	condition class	HOLD / restrict / MRM / STOP	80/250/300	<20kg	center
Dispatch/routing ECU	1	trip execution	-	fleet control	100/300/400	20–40kg	upper
ETA/latency control	1	synchronization	latency≤50ms	AD stability target	80/250/300	<20kg	upper
4 Drivetrain		electric drive	-	AD ride stability	-	-	corners
IWM	4	80kW/wheel	80×4	320kW	320/500/500	1,000–1,400kg	corners
Regen brake	4	regenerative	-	4 wheels	400/400/400	120–180kg	corners
Air brake	4	braking	-	4 wheels	400/400/400	120–180kg	corners
Knuckle+EPS	4	steer support	-	4 wheels	260/320/420	250–350kg	corners
Air suspension+CFRP	4	vibration suppression	-	4 wheels	260/350/450	250–350kg	corners
CTIS tire	4	pressure control	-	4 wheels	220/500/500	included	corners
Ride comfort target	1	no motion sickness	lateral accel control	UX stabilization	-	-	cabin
5 Structure		flat payload frame	-	550–600mm target	-	-	overall
Crossmember	6	structural beam	-	6 beams	160/1600/160	250–400kg	lateral
MH main chassis	2	H2 storage+frame	-	≈10kg-H2	400/350/5500	400–800kg	longitudinal
LH2 subframe	2	cooling+support	-	2 systems	260/450/2200	300–500kg	lower
Protection/firewall	1	impact+fire safety	-	integrated	50–100/car	400–700kg	lower/center
Container lock	4	swap fixation	-	4 points	180/300/300	80–120kg	upper
Swap positioning	4	±mm alignment	-	4 points	180/200/200	80–120kg	upper
Service interface	1	refill/drain/inspect	-	gas/LH2/water	150/500/900	<120kg	L/R
Boarding assist	1	kneel+2-step	–100mm	access assist	220/300/800	120–200kg	right
Payload margin	1	25t GVW-verified curb mass	full integration required	TBD; prior 10–13t claim removed	-	-	overall
Service margin	1	maintenance reserve	total×10–15%	reserve	-	included	overall

2035 Vehicle Platform

A 12m platform targets a 380–400mm low floor. The platform-weight and 13–16t payload claims are removed pending full integrated mass, body, fluids, passengers, wiring/piping, axle-load, and certification data. The energy balance now separates the 400kWh air-battery contribution from PCFC energy demand.

PCFC-only hydrogen demand is 48.1kg/day ($=1,200 \div 33.3 \div 0.75$). With a 400kWh air-battery contribution, net PCFC energy is 800kWh and net H₂ demand is 32.0kg/day. Six mills at 0.865kg/h for 12.5h produce 64.9kg/day, leaving a 32.9kg/day gross surplus before process losses. The 10–15kg LH₂ buffer still requires about 340L; the envelope is corrected to 350×850×1,200mm (357L). Rejected heat at 240kW and 75% efficiency is approximately 80kW.

Component	Qty	Performance	Formula	Corrected result	H/W/D mm	Weight	Layout
1 Energy/H ₂ /Power		system demand	$1.2 \times 1,000$	1,200kWh	-	-	overall
Gross H ₂ before air battery	1	PCFC 75%	$1,200 \div (33.3 \times 0.75)$	48.1kg-H ₂	-	-	overall
Air-battery energy	4	100kWh/unit	100×4	400kWh	120/350/1200	≈849kg	right
H ₂ reduction equivalent	1	25kWh/kg-H ₂	$400 \div 25$	16.0kg-H ₂ eq.	-	-	right
Net PCFC energy	1	1,200–400	-	800kWh	-	-	overall
Net H ₂ required	1	PCFC 75%	$800 \div (33.3 \times 0.75)$	32.0kg-H ₂	-	-	overall
Hydrogen generation	6	0.865kg/h	$0.865 \times 6 \times 12.5$	64.9kg-H ₂	240/450/650	720–1,080kg	F/R
Gross surplus	1	before process losses	$64.9 - 32.0$	32.9kg-H ₂	-	-	overall
LH ₂ buffer	1	10–15kg reserve	-	10–15kg-H ₂	350/850/1200	100–150kg	left
LH ₂ installed volume	1	insulation×1.6	$15 \div 70.8 \times 1.6$	≈340L	350/850/1200	included	left
Water tank	6	9kg/kg-H ₂ gross	64.9×9	584L	200/400/1200	584kg	left
Slurry tank	6	water×1.4	584×1.4	818L	230/450/1300	980–1,227kg	left
Titanium cartridge	6	3kg/0.865kg-H ₂	$64.9 \div 0.865 \times 3$	≈225kg	160/300/800	225kg	L/R
PCFC modules	6	40kW/unit	40×6	240kW	180/900/1200	720–1,080kg	upper center
PCFC efficiency	1	electrical target	-	75% target	-	-	center
AMR liquefier	2	surplus-H ₂ process	$10 \text{ kg/day} \times 2$	process target	240/450/800	250–400kg	F/R
H ₂ purifier	3	50L/min	50×3	150L/min	180/250/500	120–180kg	F/R
Gas/liquid separator	3	50L/min	50×3	150L/min	200/250/450	120–180kg	F/R
Process-water management	2	condensate / humidity balance	validated setpoint	closed-loop target	150/250/400	80–120kg	front
O ₂ management membrane	2	optional cathode control	test gate	efficiency TBD	150/250/450	80–120kg	front
CO ₂ membrane	0	direct-H ₂ baseline	not required	excluded	-	0kg	rear
Steam separator	2	H ₂ O separation	water balance	condensate recovery	150/250/450	80–120kg	rear
Reformer	0	direct-H ₂ baseline	not required	excluded	-	0kg	front
Anode loop	1	fuel recirculation	20% target	efficiency gain target	180/250/500	80–120kg	center
LH ₂ pump	2	50L/min	50×2	100L/min	150/220/350	60–100kg	left
Vaporizer	2	20kg/h	20×2	40kg/h	180/300/600	100–160kg	left
Pressure regulator	4	high→low P	-	4 systems	100/200/300	40–60kg	overall
Shutoff valve	6	safety shutoff	-	6 systems	100/180/250	40–60kg	overall
Vent line	2	safety vent	-	2 systems	100/100/1200	20–40kg	upper
Safety sensors	1	H ₂ /P/T monitor	-	integrated	80/200/300	<30kg	overall
2 Electric/Thermal		PCFC power control	-	battery transient	-	-	center
Inverter	2	120kW	120×2	240kW	150/420/650	180–260kg	upper center
DC/DC	2	100kW	100×2	200kW	130/350/500	120–180kg	upper center
Response battery	1	80kWh	-	80kWh	120/900/1600	350–500kg	upper center
HV J-BOX	1	HV distribution	-	1 system	150/350/500	40–80kg	center
EMS/VCU	1	energy+vehicle ctrl	-	1 system	100/250/300	40–80kg	center
Radiator	2	rejected heat	$240 \times (1/0.75 - 1)$	≈80kW	120/400/1200	120–200kg	front
Cooling pump	2	50L/min	50×2	100L/min	120/180/250	40–80kg	overall
HTS cooling line	1	LH ₂ cooling	-	1 system	60/60/12000	80–120kg	frame
HTC/HTS cable	1	low-loss power	-	1 system	50/50/12000	60–120kg	center→wheel
Waste-heat recovery	1	TE/TPV/Rankine	validated recovery	aux. power TBD	220/500/800	200–350kg	rear
Waste-heat utilization	1	efficiency gain	-	target only	-	-	rear
Power protection	1	insulation/fuse	-	integrated	80/200/300	<30kg	center
3 AD Control		autonomous execution	-	AD operation	-	-	upper
AD computer	1	planning/control compute	validated ODD load	performance TBD by evidence	120/400/500	40–80kg	upper
Sensor suite	1	LiDAR/camera/radar	-	360° sensing	120/500/800	40–80kg	upper
Fusion/localization	1	map+IMU+GNSS	-	position control	80/300/400	<30kg	upper
ODD/DDT monitor	1	state-dependent safety gate	condition class	HOLD / restrict / MRM / STOP	80/250/300	<20kg	center
Dispatch/routing ECU	1	trip execution	-	fleet control	100/300/400	20–40kg	upper
ETA/latency control	1	synchronization	latency≤50ms	AD stability target	80/250/300	<20kg	upper
4 HTS Drivetrain		SC drive+ride ctrl	-	no-sick UX	-	-	corners
LH ₂ SC-IWM	4	80kW/wheel	80×4	320kW	260/450/450	800–1,200kg	corners
SC regen brake	4	regen braking	-	4 wheels	300/400/400	120–180kg	corners
Air brake	4	braking	-	4 wheels	300/400/400	120–180kg	corners
Knuckle+EPS	4	steer support	-	4 wheels	220/300/400	200–300kg	corners
Air sus.+CFRP	4	vibration cut	-	4 wheels	220/300/400	200–300kg	corners
CTIS/CFRP wheel	4	pressure+light wheel	-	4 wheels	200/450/450	included	corners
Ride comfort target	1	no motion sickness	lateral accel control	UX stabilization	-	-	cabin
5 Structure		flat payload frame	-	380–400mm target	-	-	overall
HTS crossmember	6	beam+cooling	-	6 beams	120/1600/120	200–300kg	lateral
MH main chassis	2	H ₂ storage+frame	-	≈20kg-H ₂	300/300/5500	300–600kg	longitudinal
LH ₂ /HTS subframe	2	cooling+support	-	2 systems	200/400/2000	200–400kg	lower
Protection/firewall	1	impact+fire safety	-	integrated	30–80/vehicle	300–600kg	lower/center
Container lock	4	swap fixation	-	4 points	150/300/300	80–120kg	upper
Swap positioning	4	±mm alignment	-	4 points	150/200/200	80–120kg	upper
Service interface	1	refill/drain/inspect	-	water/Ti/slurry	120/400/800	<100kg	L/R
Boarding assist	1	kneel+2-step	–100mm	access assist	200/300/800	120–200kg	right
Payload margin	1	25t GVW-verified curb mass	full integration required	TBD; prior 13–16t claim removed	-	-	overall
Service margin	1	maintenance reserve	total×10–15%	reserve	-	included	overall

IRR and Cash Flow Structure

This page reconciles the cash-flow table with the corrected PL. The base case is pre-tax and unlevered: Free Cash Flow = EBITDA – CAPEX – ΔWorking Capital – Cash Taxes, with ΔWorking Capital and Cash Taxes set to zero only for this screening calculation. Depreciation is excluded from FCF. A production investment case must add tax, working capital, financing, residual value, and removal CAPEX. IRR is the discount rate that sets NPV to zero. Project payback is the first year in which cumulative nominal FCF becomes non-negative. Discounted payback is assessed separately. Node payback cannot be calculated because node-level revenue, OPEX, CAPEX, tax, and renewal cash flow are not separately specified.

Investment / Instrument	Timing	Dep.	Purpose	Cash-flow role	FCF / decision impact			
Initial Development Investment (2025–2030)								
Platform / AI / Urban OS	2025–2030	5y	Core control platform	Digital backbone	Upfront CAPEX; D&A is non-cash			
MaaS Vehicle Prototype	2025–2028	5y	Vehicle feasibility	Prototype / validation	Upfront CAPEX; no node revenue			
HTS / Battery / Mill	2026–2030	10y	Energy feasibility	Energy-system enabler	CAPEX; energy OPEX sensitivity			
Initial Nodes (4)	2027–2030	15y	Operational deployment	Initial field capacity	CAPEX; node payback required			
HPC / GTFC	2027–2030	15y	Control infrastructure	Compute / energy control	CAPEX; availability enabler			
Expansion Investment (2030–2035)								
Additional Nodes (36 total)	2030–2035	15y	City-scale deployment	Network capacity	CAPEX; utilization gate			
Fleet Expansion (500)	2030–2035	5y	Utilization scaling	Revenue capacity	CAPEX plus fleet OPEX			
Distributed Energy Infrastructure	2030–2035	10–15y	Hydrogen/electricity scaling	Energy supply	CAPEX; energy cost/trip			
AI / Dispatch Scaling	2030–2035	5y	Real-time optimization	Operating efficiency	CAPEX/OPEX; dispatch uplift			
Node Synchronization Infrastructure	2032–2035	15y	Distributed control	Network orchestration	CAPEX; node/fleet efficiency			
Corrected P/L Reference	Revenue	EBITDA	Operating Profit	EBITDA Margin	Operating Margin			
2030	¥29.481B	¥11.998B	▲¥18.208B	40.7%	-61.8%			
2035	¥131.151B	¥39.844B	▲¥53.346B	30.4%	-40.7%			
Cash-flow formulas: FCF _t = EBITDA _t – CAPEX _t – ΔWC _t – Cash Taxes _t ; DCF _t = FCF _t ÷ (1+r) ^t ; IRR solves Σ FCF _t ÷ (1+IRR) ^t = 0.								
Year	Revenue	EBITDA	CAPEX	FCF	DF 8%	Discounted FCF	Cumulative nominal FCF	Cumulative discounted FCF
2025	¥0.000B	▲¥5.000B	▲¥35.000B	▲¥40.000B	1.000	▲¥40.000B	▲¥40.000B	▲¥40.000B
2026	¥0.000B	▲¥6.000B	▲¥40.000B	▲¥46.000B	0.926	▲¥42.593B	▲¥86.000B	▲¥82.593B
2027	¥2.000B	▲¥5.000B	▲¥45.000B	▲¥50.000B	0.857	▲¥42.867B	▲¥136.000B	▲¥125.460B
2028	¥8.000B	▲¥2.000B	▲¥40.000B	▲¥42.000B	0.794	▲¥33.341B	▲¥178.000B	▲¥158.800B
2029	¥18.000B	¥3.000B	▲¥35.000B	▲¥32.000B	0.735	▲¥23.521B	▲¥210.000B	▲¥182.321B
2030	¥29.481B	¥11.998B	▲¥28.000B	▲¥16.002B	0.681	▲¥10.891B	▲¥226.002B	▲¥193.212B
2031	¥55.000B	¥28.000B	▲¥22.000B	¥6.000B	0.630	¥3.781B	▲¥220.002B	▲¥189.431B
2032	¥95.000B	¥52.000B	▲¥20.000B	¥32.000B	0.583	¥18.672B	▲¥188.002B	▲¥170.759B
2033	¥145.000B	¥82.000B	▲¥18.000B	¥64.000B	0.540	¥34.577B	▲¥124.002B	▲¥136.182B
2034	¥198.000B	¥118.000B	▲¥15.000B	¥103.000B	0.500	¥51.526B	▲¥21.002B	▲¥84.657B
2035	¥131.151B	¥39.844B	▲¥12.000B	¥27.844B	0.463	¥12.897B	¥6.842B	▲¥71.759B
Financial Result / Value Gap	Value	Interpretation	Assumption / Test	Definition	Decision meaning			
Total CAPEX 2025–2035	¥310.000B	Sum of annual CAPEX	Model basis	Pre-tax / unlevered	Screening model			
Project IRR	0.47%	Below 12% hurdle	Discount rate	8%	NPV and discounted payback			
NPV @ 8%	▲¥71.759B	Base-case value gap	ΔWorking capital	0	Screening only			
NPV @ 12%	▲¥90.771B	Hurdle-rate value gap	Cash taxes	0	Screening only			
Nominal project payback	2035	Cumulative nominal FCF ≥ 0	Depreciation	Excluded from FCF	Retained in P/L			
Discounted payback	Not achieved	Cumulative DCF < 0	Financing / residual / salvage	Excluded	Separate scenarios required			
Node payback	Not calculated	Node-level cash flow absent	Removal CAPEX	Excluded	Add if applicable			
2035 EBITDA margin	30.4%	Cash generation before D&A	NPV@8% break-even	+¥71.759B PV	Close present-value gap			
2035 Operating margin	-40.7%	Depreciation-heavy loss	Equivalent FCF uplift	+¥21.118B/y (2030–35) or +¥154.923B in 2035	Redesign assumptions / stage CAPEX			
Operational KPI	Financial driver	Impact on FCF	Sensitivity / gap test	Explicit change	Decision use			
Utilization	Revenue / fixed-cost absorption	↑ monetized hours → ↑ EBITDA / FCF	Base case	As displayed	IRR 0.47%; NPV@8% ▲¥71.759B			
Fleet idle ratio	OPEX / CAPEX	↓ idle hours → ↓ cost / deferred fleet CAPEX	NPV@8% break-even	+¥21.118B FCF/y in 2030–35	Minimum equal annual uplift			
Matching + dispatch	Revenue	↑ realized trips → ↑ revenue	Terminal-only equivalent	+¥154.923B FCF in 2035	High-risk concentration of value			
Energy cost/trip	OPEX / margin	↓ unit energy cost → ↑ EBITDA	Hurdle-rate gap	+¥90.771B PV at 12%	12% investment gate			
Node utilization	Revenue / CAPEX	↑ revenue per installed node → ↑ ROIC	CAPEX-only equivalent	~30.5% PV-weighted CAPEX reduction	If EBITDA is unchanged			
Load factor / price	Revenue efficiency	↑ contribution per trip → ↑ FCF	Required model set	Revenue, OPEX, CAPEX, WC/tax, financing, residual/removal	Re-run base / upside / downside			
Sensitivity case	Explicit cash-flow change	IRR	NPV @8%	NPV @12%	Interpretation			
Base case	As displayed	0.47%	▲¥71.759B	▲¥90.771B	Fails both screening and hurdle gates			
FCF uplift 1	+¥10.000B/y in 2030–2035	4.28%	▲¥37.780B	▲¥64.642B	Improves value but remains negative			
FCF uplift 2	+¥20.000B/y in 2030–2035	7.65%	▲¥3.800B	▲¥38.514B	Near 8% break-even; below hurdle			
NPV@8% break-even	+¥21.118B/y in 2030–2035	8.00%	¥0.000B	▲¥35.591B	Clears 8% only			
CAPEX reduction	-20% each year; EBITDA unchanged	5.15%	▲¥24.665B	▲¥48.786B	Insufficient alone			
12% hurdle case	+¥281.921B terminal FCF in 2035	12.00%	¥58.825B	¥0.000B	Equivalent hurdle closure; not a forecast			
Cash-flow completeness item	Current treatment	Required production-case addition			Financial effect			
Working capital	0	Receivables, payables, inventory and timing by year			FCF / payback			
Cash tax	0	Tax loss carry-forward, tax rate, incentives and timing			FCF / NPV			
Financing	Excluded	Debt/equity, interest, fees, repayment and covenants			Equity IRR / DSCR			
Residual / salvage	Excluded	Terminal asset value by asset class			Terminal FCF / NPV			
Removal / decommissioning	Excluded	End-of-life removal, restoration and disposal cash flow			Terminal FCF			
Maintenance / renewal	Not separately shown	Overhaul, replacement and lifecycle-maintenance cadence			OPEX / CAPEX			
Node cash flow	Not separately shown	Revenue, OPEX, CAPEX, tax and renewal by node / phase			Node payback / ROIC			
Inflation / FX / nominal-real basis	Not explicit	Define price basis, currency, escalation and conversion			Comparability / risk			

Financial Interpretation & KPI Linkage

The corrected base case becomes nominally cash-positive in 2035, but the project IRR is 0.47% and NPV at 8% is negative (▲¥71.759B). It therefore does not satisfy the 12% investment hurdle. This is a decision result, not a calculation error: the business assumptions must be revised or separately validated before claiming investment feasibility.

The largest unresolved drivers are node monetization and delivery cost, fleet operating cost, hydrogen volume/cost, platform pricing, depreciation-heavy development investment, working capital, tax, and renewal CAPEX. KPI improvements affect finance only through explicit volume, price, cost, cash, and investment equations.

Node Deployment Logic	Node count	What the phase validates		Objective / Exit evidence
Phase 1 (2027–2030)	4	Initial AVP / MaaS and node economics		Demand signal, operational feasibility, energy supply, permits, initial node cash flow
Phase 2 (2030–2032)	12	Fleet synchronization and repeatability		Repeatable dispatch, uptime, supply, SLA, maintenance, node payback evidence
Phase 3 (2032–2035)	36	City-scale operation and network economics		Cross-node utilization, revenue scale, resilience, auditability, staged investment readiness

KPI	Financial impact	Diagnosed symptom	Diagnostic question (KPI → P/L)	Priority
Utilization	Revenue / fixed-cost absorption	Stranded vehicle-hours	Are monetized vehicle-hours sufficient to recover fleet and node fixed cost?	Critical
Fleet idle ratio	OPEX / CAPEX efficiency	Idle assets / over-fleet	Can repositioning or fleet deferral reduce idle cost and future CAPEX?	Critical
Matching rate	Trips / revenue	Demand not realized	Are spatial, temporal, service, or price constraints preventing matching?	High
Dispatch success	Revenue stability	Matched trips lost	Is fleet allocation, SLA, routing, or recovery logic causing failed dispatch?	High
Energy cost/trip	Margin / FCF	Unit-cost creep	Does metered end-to-end energy cost support ≤¥500 / ≤¥350 per trip?	Critical
Node utilization	CAPEX efficiency	Underused nodes	Is revenue per installed node above the node-level hurdle rate?	Critical
Supply stability	Revenue continuity	Service hours lost	Is supplied energy + required energy ≥95% after losses and downtime?	Critical
Load factor	Revenue efficiency	Low seat/cargo monetization	Does service mix, ridesharing, or freight integration improve revenue per trip?	High
CAPEX intensity	IRR / NPV	High upfront outflow	Can phasing, leasing, asset sharing, or design simplification reduce CAPEX?	Critical
ODD / DDT availability	Utilization / revenue	Unavailable service time	Does validated safety, fallback, remote support and ODD evidence expand sellable hours?	Critical

Contradiction / Completion Check	Expected	Status	Observation / Required action
P/L formulas and totals reconciled	YES	PASS	Displayed formulas and totals re-summed.
H2 revenue and COGS volumes aligned	YES	PASS	Revenue and cost must use the same physical volume basis.
Depreciation rows re-summed	YES	PASS	D&A is retained in P/L but excluded from FCF.
FCF tied to displayed EBITDA and CAPEX	YES	PASS	Annual cash-flow table reconciled.
IRR recalculated from displayed FCF	YES	PASS	Project IRR = 0.47%.
NPV at 8% calculated	≥0	FAIL	NPV = ▲¥71.759B.
NPV at 12% calculated	≥0	FAIL	NPV = ▲¥90.771B at the hurdle rate.
Nominal project payback calculated	YES	PASS	First non-negative cumulative FCF occurs in 2035.
Discounted payback calculated	By 2035	FAIL	Cumulative discounted FCF remains ▲¥71.759B.
Node payback separately calculated	YES	TBD	Node revenue, OPEX, CAPEX, tax and renewal cash flows are absent.
Working capital / tax / residual / removal	Included	TBD	Omitted from the screening case; add separate scenarios.
Field KPI evidence measured	YES	TBD	Utilization, dispatch, energy, supply and node targets are not measured outcomes.
ODD / DDT / safety-case evidence	Approved	TBD	Operational evidence is not shown in the P/L.
Revenue double-counting eliminated	YES	REVIEW	Base fares, platform fees, node fees and settlements require contract-level separation.
Word / PDF visual verification	YES	PASS	Target pages rendered and reviewed after editing.

Decision Rules & Gating Conditions INVEST only if all critical gates pass. STAGE when failures are fixable in the next phase. HOLD / REWORK when critical evidence is missing. DO NOT SCALE when NPV < 0 and hurdle IRR fails. G1 IRR ≥12%: 0.47% — FAIL G2 NPV@8% ≥0: ▲¥71.759B — FAIL G3 NPV@12% ≥0: ▲¥90.771B — FAIL G4 Nominal payback by 2035: 2035 — PASS G5 Discounted payback by 2035: not achieved — FAIL G6 Node payback ≤9y: not calculated — TBD	Logic Summary / Required Enhancements BASE-CASE DECISION: STAGE / REWORK Do not invest at scale. Advance only through gated pilots until node economics, KPI evidence and value gaps are closed. Required enhancements: • Higher verified revenue, load and price • Lower CAPEX/OPEX through phasing and asset sharing • Separate node-level cash flow and payback • Meter energy cost, losses, uptime and maintenance • Add WC, tax, financing, residual and removal cash flows • Re-run base / upside / downside cases	Missing Evidence / Actions Required Profitability remains dominated by utilization, load, price, energy, fleet OPEX and infrastructure CAPEX. Node conversion requires contract, COGS, utilization, permit and residual-value evidence. AVP sharing may support private users and MaaS fleets, but revenue cannot be recognized twice. Hydrogen integration reduces OPEX only when metered end-to-end cost and uptime support the claim. OEMs, AD platforms, mobility platforms, operators, suppliers and finance providers need explicit RACI and contracts.
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KPI / Metric	2030 target	2035 target	Base case / input	Status	Evidence / interpretation
Utilization	0.75	0.85	Design target	VALIDATE	Primary volume driver; field evidence required.
Fleet idle ratio	0.25	0.15	Design target	VALIDATE	OPEX/CAPEX efficiency; measured idle hours required.
Energy cost/trip	≤¥500	≤¥350	Design target	VALIDATE	Meter production, storage, loss, uptime and maintenance.
Dispatch success	≥95%	≥99%	Design target	VALIDATE	Revenue stability; SLA and recovery data required.
Matching rate	≥90%	≥95%	Design target	VALIDATE	Demand realization; segment by service/time/area.
Node utilization	≥80%	≥90%	Design target	VALIDATE	CAPEX efficiency; calculate revenue per installed node.
Trips/day	73	140	Scenario	SCENARIO	Revenue-scale assumption, not measured outcome.
Load factor	0.70	0.80	Scenario	SCENARIO	Revenue efficiency assumption.
EBITDA margin	40.7%	30.4%	Calculated	MIXED	Cash margin remains positive but declines by 2035.
Operating margin	-61.8%	-40.7%	Calculated	FAIL	Accounting loss remains negative after D&A.
Project IRR	≥12%	≥12%	0.47%	FAIL	Below investment hurdle.
NPV @8% / @12%	≥0	≥0	▲¥71.759B / ▲¥90.771B	FAIL	Negative at both screening and hurdle rates.
Nominal payback	By 2035	By 2035	2035	PASS	Achieved only at the end of the horizon.
Discounted payback	By 2035	By 2035	Not achieved	FAIL	Cumulative DCF remains negative.
Node payback	≤9y	≤9y	Not calculated	TBD	Node-level cash flows must be separated.
Latency / intervention / supply	≤100ms	Tighten	Design targets	VALIDATE	Must be field-validated and linked to sellable hours.

Evidence domain	Missing evidence	Phase 1 gate (4 nodes)	Phase 2/3 gate (12–36)	Decision consequence
Demand / revenue	Anchor contracts, price/take-rate, utilization and load evidence	Paid demand and initial margin	Repeatable volume, price and contribution	No scale without monetized demand
Fleet / operations	Dispatch, idle, recovery, maintenance and service-SLA data	Stable pilot operations	Multi-node repeatability and recovery	Fleet sizing and OPEX gate
Energy	Metered cost/trip, losses, storage, uptime and supply stability	≤¥500/trip; supply ≥95%	≤¥350/trip; higher resilience	Energy OPEX viability
Node economics	Site, permit, CAPEX, OPEX, revenue, tax and renewal cash flows	Node-level P/L and payback	Repeatable network payback	Stage or stop node CAPEX
Safety / ODD / DDT	Safety case, latency, intervention, fallback and remote-support evidence	Approved initial ODD	Evidence-led ODD expansion	No sellable hours outside validated ODD
Finance	Working capital, tax, financing, residual, salvage and removal cash flows	Complete screening model	Auditable investment case	IRR / NPV / payback gate
Commercial / RACI	Revenue recognition, liability, authority, contracts and settlements	Signed responsibility matrix	Scalable operating contracts	Prevent double counting and accountability gaps
Independent review	Technical, legal, insurance and financial validation	Gate review before Phase 2	Investment-committee review before scale	Proceed / hold / rework decision

Base-case decision remains STAGE / REWORK until all critical evidence gates pass; do not invest at scale on the current assumptions.

Design Constraints & System Physics - Corrected Source Register

Values are classified as Official value, Technical target, Scenario assumption, or Reference only. A source link does not convert a scenario assumption into a fact. Rows with no page-specific or value-specific support are retained as references but must not be used as validated inputs.

Category	Variable	Corrected value / status	Source	Evidence classification / use
Constraint	Insurance requirement reference	Defined only after exact provision identified	MLIT guidance	Reference only
	Operational-time regulation	Define by service/vehicle/operator law	MLIT transport	Reference only
	Safety-margin requirement	Safety-case derived; no generic numeric value	MLIT safety	Reference only
	Traffic regulation	Applicable rules must be mapped to ODD	NPA self-driving	Official framework
	Weather constraint	ODD-specific threshold required	MLIT AD	Reference only
Cost	Charging infrastructure cost	10–50M JPY/unit retained as scenario range	IEA EV Outlook	Scenario assumption
	EV cost benchmark	5–10M JPY/unit retained as scenario range	IEA EV Outlook	Scenario assumption
	Hydrogen-station CAPEX	150–300M JPY retained as scenario range	IEA Future of H2	Scenario assumption
	Software cost ratio	10–30% retained as scenario range	OECD finance	Scenario assumption; not source-specific
Demand	Aichi population	Use latest prefectural table before submission	Statistics Bureau	Official table; value not repeated here
	Japan population	122.68M at 2026-08-01	STAT POP 202608	Official value
	Taxi fare	Local fare table required; no national 2,000–2,500 constant	MLIT taxi	Official framework
	Trips per person	2.3/day retained only as survey assumption	MLIT transport survey	Scenario assumption
	Vehicle stock in Aichi	Use latest Chubu table	Chubu vehicle stats	Official table; validate date
Energy	Battery storage cost trend	Declining; exact year/system boundary required	NREL battery	Reference only
	Electricity intraday price	13.0 JPY/kWh retained only with period definition	METI ELEC p14	Scenario/period-specific
	Electricity retail price	30.2 JPY/kWh	METI ELEC p14	Official cited benchmark
	Electricity spot price	12.24 JPY/kWh retained only with period definition	METI ELEC p14	Period-specific
	Hydrogen LHV	33.3 kWh/kg	DOE H2 LHV	Physical constant
	Hydrogen pathway roundtrip efficiency	Lower than direct electricity; system boundary required	NREL H2 pathway	Reference only
	Hydrogen production cost	150–200 JPY/kg retained as scenario	IEA Future of H2	Scenario assumption
	Hydrogen sales price	300–400 JPY/kg retained as scenario	NEDO H2 library	Scenario assumption
	Hydrogen specific energy	120 MJ/kg	DOE H2 LHV	Physical constant
	Hydrogen storage cost	DOE 2020 target: \$333/kg-H2 stored	DOE 2020 STORAGE	Technical target; not achieved current value
	Hydrogen storage gravimetric	DOE 2020 target: 4.5 wt%	DOE 2020 STORAGE	Technical target; not achieved current value
	PEMFC direct-H2 efficiency	~60% LHV	DOE PEMFC 60	Technology comparison; not 22%
	LNG energy density	13.9 kWh/kg retained	IEA Future of H2	Reference; page/definition required
	LNG volumetric energy	20.8 MJ/L retained	IEA Future of H2	Reference; page/definition required
	Liquid-H2 volumetric energy	~8 MJ/L retained	DOE H2 storage	Reference
	Generation cost benchmark	LCOE reference only	IEA generation cost	Reference only
	PV high-output benchmark	>1,100kWh/kW/yr	NEDO PV p14	Benchmark
	PV output benchmark	1,000–1,100kWh/kW/yr	NEDO PV p14	Benchmark
	SOFC efficiency	~60% scenario benchmark	DOE fuel cells	Technology-dependent
Financial	Discount rate	8–12% retained as screening range	OECD finance	Scenario assumption
	Infrastructure return	Project-specific	JBIC	Reference only
	Investment hurdle	Project-specific	OECD finance	Reference only
	Project IRR benchmark	10–15% retained as hurdle range	JBIC	Scenario assumption
Infra	Charging infrastructure	Definition/reference only	IEA EV Outlook	Reference only
	Energy distribution	Definition/reference only	IEA generation cost	Reference only
	Hydrogen infrastructure	Definition/reference only	IEA Future of H2	Reference only
Logistics	Standard freight fare	Distance/time and vehicle-class tables; no national 52 JPY/kg constant	MLIT STANDARD FARE	Official framework
Market	Global EV sales share	15% retained only with year/definition	IEA EV Outlook	Period-specific
	Heavy-truck new sales (Japan)	73k retained only after exact JAMA table validation	JAMA statistics	Validate year/category
	Japan EV sales share	1% retained only with year/definition	IEA EV Outlook	Period-specific
	Total new vehicle sales (Japan)	4.421M retained only after exact JAMA table validation	JAMA statistics	Validate year/category
Policy	Japan 2035 electrified-vehicle target	100% new passenger-car sales; scope definition required	METI 2035 policy	Policy definition required
	Electrified-vehicle definition	BEV/PHEV/HEV/FCEV	METI EV scope	Policy definition
	Powertrain-mix reference	Use dated industry chart	JAMA powertrain mix	Reference only
Regulation	Accident response obligation	Required	NPA L4	Official
	Administrative cancellation	Applicable	NPA L4	Official
	Autonomous-driving subsidy availability	Program-specific	MLIT subsidy	Verify FY/eligible costs
	Current safety-policy baseline	Applicable	MLIT safety	Official framework
	Government safety policy	Defined	MLIT AD	Official framework
	Human-operated-vigilance requirement	Depends on operation type	NPA L4	Official framework
	Infrastructure regulation	Defined by installed equipment/site	MLIT AD	Map exact law
	Insurance requirement	Required; exact coverage required	MLIT AD	Map exact provision
	International standards relevance	Relevant, not automatically legally binding	MLIT safety	Reference
	Legal-liability framework	Defined across product/operator/road law	MLIT safety	Map exact provisions
	Level 4 summary availability	Available	NPA L4	Official
	Local-government consultation	Required where specified	NPA L4	Official
	Monitoring requirement	Required where specified	NPA L4	Official
	Non-specific operation treatment	Defined	NPA interpretation	Official interpretation
	ODD requirement	Required	NPA L4	Official
	Operation plan submission	Required	NPA portal	Official
	Passenger transport rule	Applicable by service model	e-Gov Passenger Rule	Official law
	Permit cancellation	Applicable	NPA L4	Official
	Police reporting	Required where specified	NPA L4	Official
	Public-road operation framework	Defined	MLIT AD	Official
	Remote monitoring	Allowed under defined conditions	NPA L4	Official
	Road Traffic Act	Applicable	e-Gov Road Traffic Act	Official law
	Road Transport Vehicle Act	Applicable	e-Gov Vehicle Act	Official law
	Road-use regulation	Defined	MLIT safety	Official
	Running-environment condition	Required	MLIT AD	Official
	Specific autonomous-operation permit	Required	NPA portal	Official
	Subsidy eligible cost	Program-specific	MLIT subsidy	Verify current notice
	Suspension authority	Applicable	NPA L4	Official
	Transport safety law reference	Defined	MLIT safety	Official
	Vehicle Act enforcement order	Applicable	e-Gov order	Official law

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